

A note on connections between the Föllmer process and the denoising diffusion probabilistic model

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Abstract

The Föllmer process is a Brownian motion conditioned to have a pre-specified distribution at time 1. This process can be interpreted as an “augmented” time-compressed version of the reverse stochastic differential equation (SDE) for the denoising diffusion probabilistic model (DDPM). While this fact has been indirectly used to analyze DDPM sampling errors via discretization of the reverse SDE, connections between direct discretization of the Föllmer process and the DDPM sampler have not yet been fully explored. This note aims to clarify this point while surveying relevant results from existing work. We show that discretized Föllmer processes give natural hyper-parameter settings of the DDPM sampler. Moreover, this allows us to systematically recover state-of-the-art results on DDPM sampling error bounds with slight improvements.

Keywords: diffusion model, discretization, low-dimensional adaptation, sampling error, score-based generative modeling, stochastic localization.

1 Introduction

Let μ be a probability distribution on $\mathbb{R}^d$. Following [17], we call a d -dimensional continuous stochastic process $X = (X_t)_{t \in [0,1]}$ the *Föllmer process* associated with μ if $X_1 \sim \mu$ and the conditional law of X given X_1 is the d -dimensional Brownian bridge from 0 to X_1 on $[0, 1]$. This process has recently received renewed attention in various fields of both theory and application such as studies of functional inequalities [16, 17, 18, 36, 38, 45, 46] as well as high-dimensional central limit theorems [19, 21, 22], optimization [9, 43], generative modeling [5, 59], sampling from probability distributions [20, 30, 55, 61] and Bayesian inference [52, 57]. Historically, the Föllmer process appeared as the solution to an entropy minimization problem posed by Schrödinger. A modern formulation of Schrödinger’s problem is as follows (see [39, 44] for more details). Let $\mathbb{W}^d := C([0, 1], \mathbb{R}^d)$ be the space of continuous functions from $[0, 1]$ to $\mathbb{R}^d$ equipped with the uniform topology. For a probability distribution Q on $\mathbb{W}^d$, we denote by Q_0 and Q_1 the image measures of Q by the maps $w \mapsto w(0)$ and $w \mapsto w(1)$, respectively. Then, given a pair (μ_0, μ_1) of probability distributions on $\mathbb{R}^d$ and a probability distribution R on $\mathbb{W}^d$, the *dynamical Schrödinger problem* associated with the reference measure R is the following minimization problem with respect to probability

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distributions Q on $\mathbb{W}^d$:

$$\min_{Q: Q_0=\mu_0, Q_1=\mu_1} H(Q \mid R). \quad (1.1)$$

Here, given two probability measures P and Q defined on a common measurable space, $H(P \mid Q)$ denotes the relative entropy of P with respect to Q , i.e.

$$H(P \mid Q) = \int \frac{dP}{dQ} \log \left(\frac{dP}{dQ} \right) dQ \quad \text{if } P \ll Q$$

and $H(P \mid Q) = \infty$ otherwise. Föllmer [25] showed that when $H(\mu \mid \gamma_d) < \infty$ with $\gamma_d = N(0, I_d)$ the d -dimensional standard normal distribution, the law of X gives the unique solution to (1.1) with $\mu_1 = \mu$, μ_0 being the unit mass at 0 and R being the d -dimensional Wiener measure; see Section II.1.3 *ibidem*.

Under the finite entropy condition $H(\mu \mid \gamma_d) < \infty$, it follows from Föllmer [24]’s results that X is a weak solution to the following stochastic differential equation (SDE):

$$dX_t = \left(\frac{X_t}{t} + \nabla \log p_t(X_t) \right) dt + dW_t \quad (0 \leq t \leq 1), \quad X_0 = 0, \quad (1.2)$$

where $W = (W_t)_{t \in [0,1]}$ is a standard Wiener process in $\mathbb{R}^d$ and for every $0 < t < 1$, p_t denotes the density of X_t , i.e.

$$p_t(x) = \frac{1}{\{2\pi t(1-t)\}^{d/2}} \int_{\mathbb{R}^d} \exp \left(-\frac{|x - ty|^2}{2t(1-t)} \right) \mu(dy), \quad x \in \mathbb{R}^d.$$

See Remark 2.2 for details and also [10, 33] for related results. Indeed, we will show that X always solves (1.2) without *any* assumption on μ ; see Theorem 2.1. $\nabla \log p_t$ is known as the *score function* of (the law of) X_t , and the form of (1.2) reminds us of the SDE formulation of *score-based generative modeling* (SGM) in Song *et al.* [54]. In fact, it turns out that X is an “augmented” time-compressed version of the reverse SDE corresponding to the *denoising diffusion probabilistic model* (DDPM) of [29]; see Proposition 3.1. Although this fact is essentially known in the literature (see the discussion preceding Proposition 3.1), existing work does not seem to fully explore connections between discretization of the Föllmer process and the DDPM sampler. This note aims to clarify this point while surveying relevant results in the literature. As we will see in Section 3, discretized Föllmer processes give natural hyper-parameter settings of the DDPM sampler. In addition, we can systematically recover state-of-the-art results on DDPM sampling error bounds in [3, 8, 31], along with slight improvements. The key to our development is the fact that the Föllmer process is strongly linked to some information-theoretic quantities and Eldan [15]’s stochastic localization. Indeed, these links were implicitly used in [3, 31] to analyze discretization errors of the reverse SDE in SGM. Our development complements these works.

While the Föllmer process has been already used for generative modeling as mentioned above, most studies have used it for developing an alternative model to the DDPM rather than analyzing the DDPM itself. In fact, Chen *et al.* [5] and Wang *et al.* [59] employ the Föllmer process as a part of more sophisticated generative models proposed therein. We also remark that a solution to the problem (1.1) in the general case, known as the *Schrödinger bridge*, is extensively employed for generative modeling. Among others, its

connection to SGM is discussed in [4, 6, 12, 49]. However, the focus of these studies is to develop a more general framework than SGM.¹

The remainder of this note is organized as follows. Section 2 reviews basic properties of the Föllmer process. Section 3 discusses its connection to the DDPM, with emphasis on its implication to sampling error evaluation. The appendix presents additional mathematical proofs.

Notation For a vector $x \in \mathbb{R}^d$, we denote the Euclidean norm of x by $|x|$. If ξ is a random vector in $\mathbb{R}^d$, P^ξ denotes the law of ξ . For a positive definite symmetric matrix Σ , we write ϕ_Σ for the density of $N(0, \Sigma)$. Given a d -dimensional stochastic process $X = (X_t)_{t \in I}$ with I an interval in $[0, \infty)$, we denote by $\mathbf{F}^X = (\mathcal{F}_t^X)_{t \in I}$ the usual augmentation of the filtration generated by X with respect to the underlying probability measure; see [35, Definition 2.7.2] or [51, p.45] for the precise definition. If $I = [0, 1]$ and X is continuous, we denote by P^X the path measure of X induced on $\mathbb{W}^d$. In particular, P^W denotes the d -dimensional Wiener measure.

2 Properties of the Föllmer process

In this section, we review basic properties of the Föllmer process. Although most results are known in the literature at least under additional regularity on μ , we give their proofs to avoid such extra assumptions.

To fix the terminology, we define the Föllmer process as follows:

Definition 2.1 (Föllmer process). For $x, y \in \mathbb{R}^d$, we write P_x^y for the law of the Brownian bridge from x to y induced on $\mathbb{W}^d$. A d -dimensional continuous stochastic process $X = (X_t)_{t \in [0, 1]}$ is called a *Föllmer process* associated with μ if its law induced on $\mathbb{W}^d$ is equal to $\int_{\mathbb{R}^d} P_0^y \mu(dy)$.

We can realize a Föllmer process associated with μ as $X_t = t\xi + \beta_t$, $t \in [0, 1]$, where $\xi \sim \mu$ and $\beta = (\beta_t)_{t \in [0, 1]}$ is a standard Brownian bridge on $[0, 1]$ independent of ξ . The next proposition shows that any Föllmer process is realized in this way:

Proposition 2.1. *Let $X = (X_t)_{t \in [0, 1]}$ be a Föllmer process associated with μ . Define a process $\beta = (\beta_t)_{t \in [0, 1]}$ as $\beta_t = X_t - tX_1$, $t \in [0, 1]$. Then β is a standard Brownian bridge independent of X_1 .*

Proof. By definition, the conditional law of β given X_1 is a standard Brownian bridge in $\mathbb{R}^d$. Since this conditional law does not depend on X_1 , the unconditional law of β is also a standard Brownian bridge and β is independent of X_1 . $\square$

In the remainder of this note, $X = (X_t)_{t \in [0, 1]}$ always denotes a Föllmer process associated with μ and $\beta = (\beta_t)_{t \in [0, 1]}$ is defined as in Proposition 2.1 unless otherwise is stated.

Proposition 2.2. *X is a Markov process.*

Proof. Since the law of the time reversal process $\tilde{X} = (X_{1-t})_{t \in [0, 1]}$ is $\int_{\mathbb{R}^d} P_x^0 \mu(dx)$, $\tilde{X}$ is a Markov process with the same transition function as the Brownian bridge and initial distribution μ . Then, by property (iii) in

¹Note that Chen & Vanden-Eijnden [6] use the term “Föllmer process” to indicate the Schrödinger bridge whose initial marginal μ_0 is the Dirac measure.

[7, Section 1.1], we have $P(A \mid \mathcal{F}_t^X) = P(A \mid X_t)$ for any $t \in [0, 1]$ and $A \in \mathcal{F}_t^{\tilde{X}}$. Since $\{X_s \in B\} \in \mathcal{F}_t^{\tilde{X}}$ for any $s \in [t, 1]$ and Borel set $B \subset \mathbb{R}^d$, we obtain $P(X_s \in B \mid \mathcal{F}_t^X) = P(X_s \in B \mid X_t)$. Hence X is a Markov process. $\square$

Remark 2.1. When $H(\mu \mid \gamma_d) < \infty$, the Markov property of X is a consequence of the fact that X is the solution to the dynamical Schrödinger problem (1.1); see Theorem 3.43 in [26] or Proposition 5 in [39].

As announced, X always solves SDE (1.2) in the weak sense without any assumption on μ :

Theorem 2.1. X is a weak solution to (1.2). More precisely, we have

$$P \left(\int_0^1 \left| \frac{X_t}{t} + \nabla \log p_t(X_t) \right| dt < \infty \right) = 1$$

and the process

$$W_t^X := X_t - \int_0^t \left(\frac{X_s}{s} + \nabla \log p_s(X_s) \right) ds, \quad t \in [0, 1],$$

is a standard $\mathbf{F}^X$ -Wiener process in $\mathbb{R}^d$.

Proof. See Section A.1. $\square$

Remark 2.2. When $H(\mu \mid \gamma_d) < \infty$, the claim of Theorem 2.1 follows from Föllmer [24]’s results in the following way: Since $H(P^X \mid P^W) = H(\mu \mid \gamma_d) < \infty$ and X is a Markov process, there exist Borel functions $b : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\hat{b} : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that both $(X_t - \int_0^t b(s, X_s) ds)_{t \in [0, 1]}$ and $(X_{1-t} - \int_0^t \hat{b}(s, X_{1-s}) ds)_{t \in [0, 1]}$ are Wiener processes in $\mathbb{R}^d$ and $\nabla \log p_t(x) = \hat{b}(1-t, x) + b(t, x)$ for almost all $t \in (0, 1)$ and $x \in \mathbb{R}^d$ by Proposition 2.11 and Theorem 3.10 in [24]. However, since $(X_{1-t})_{t \in [0, 1]}$ is a Brownian bridge with initial distribution μ and terminal value 0, we have $\hat{b}(t, x) = -x/(1-t)$. Hence we obtain $b(t, x) = x/t + \nabla \log p_t(x)$.

Next we focus on the drift process of SDE (1.2):

$$v_t := \frac{X_t}{t} + \nabla \log p_t(X_t), \quad 0 < t < 1.$$

The process $(v_t)_{t \in (0, 1)}$ is sometimes called the *Föllmer drift* associated with μ . We begin by discussing its relation to information-theoretic quantities.

Proposition 2.3 (Föllmer [24], Proposition 2.11).

$$H(\mu \mid \gamma_d) = \frac{1}{2} \int_0^1 \mathbb{E}[|v_t|^2] dt. \quad (2.1)$$

Strictly speaking, Föllmer [24] showed (2.1) when $\mu \ll \gamma_d$ (see also [38, Theorem 2]), but extension to the case $\mu \not\ll \gamma_d$ is straightforward. We give a proof in Section A.2 for the shake of completeness.

To state the next result, we need to introduce some notions. When a probability distribution ν on $\mathbb{R}^d$ has an almost everywhere differentiable density f with respect to γ_d , we define the *relative Fisher information* of ν with respect to γ_d as

$$I(\nu \mid \gamma_d) := \int_{\mathbb{R}^d} |\nabla \log f|^2 d\nu = \int_{\mathbb{R}^d} \frac{|\nabla f|^2}{f} d\gamma_d.$$

Also, for $t \in [0, 1]$, we denote by $\mathcal{S}_t\mu$ the law of a random vector $\sqrt{t}\xi + \sqrt{1-t}Z$ with $\xi \sim \mu$ and $Z \sim \gamma_d$ being independent. $\mathcal{S}_t\mu$ is known as the *Slepian interpolation* between μ and γ_d . Observe that $X_t/\sqrt{t} \sim \mathcal{S}_t\mu$ when $t > 0$. Also, when $t < 1$, $\mathcal{S}_t\mu$ has a smooth density with respect to γ_d and we denote it by f_t .

Proposition 2.4. *For every $t \in (0, 1)$, we have $v_t = \nabla \log f_t(X_t/\sqrt{t})/\sqrt{t}$. Consequently,*

$$\mathbb{E}[|v_t|^2] = \frac{I(\mathcal{S}_t\mu \mid \gamma_d)}{t}.$$

Proof. For any $t \in (0, 1)$ and $y \in \mathbb{R}^d$, we can rewrite $f_t(y)$ as

$$f_t(y) = \phi_{I_d}(y)^{-1} \int_{\mathbb{R}^d} \phi_{(1-t)I_d}(y - \sqrt{t}x) \mu(dx) = \phi_{I_d}(y)^{-1} \frac{p_t(\sqrt{t}y)}{t^{d/2}}.$$

Hence we have $\nabla \log f_t(y) = y + \sqrt{t} \nabla \log p_t(\sqrt{t}y)$. Thus $v_t = \nabla \log f_t(X_t/\sqrt{t})/\sqrt{t}$. $\square$

Combining Propositions 2.3 and 2.4 gives a version of *de Bruijn's identity* (cf. Proposition 5.2.2 in [2]):

$$H(\mu \mid \gamma_d) = \int_0^1 \frac{I(\mathcal{S}_t\mu \mid \gamma_d)}{2t} dt. \quad (2.2)$$

To obtain another useful representation of the Föllmer drift, we introduce a family of probability distributions on $\mathbb{R}^d$. For $t \in (0, 1)$ and $x \in \mathbb{R}^d$, define a probability distribution $\mu_{t,x}$ on $\mathbb{R}^d$ as

$$\mu_{t,x}(dy) = \frac{e^{\frac{|y|^2}{2} - \frac{|y-x|^2}{2(1-t)}}}{\int_{\mathbb{R}^d} e^{\frac{|z|^2}{2} - \frac{|z-x|^2}{2(1-t)}} \mu(dz)} \mu(dy).$$

Since $y \mapsto |y|^p e^{\frac{|y|^2}{2} - \frac{|y-x|^2}{2(1-t)}}$ is bounded for all $p > 0$, $\mu_{t,x}$ has all moments. We particularly set

$$m(t, x) := \int_{\mathbb{R}^d} y \mu_{t,x}(dy).$$

A straightforward computation shows

$$v_t = \frac{m(t, X_t) - X_t}{1-t} \quad \text{for all } 0 < t < 1. \quad (2.3)$$

Proposition 2.5 (Eldan & Lee [16], Eq.(21)). *For any $t \in (0, 1)$ and $x \in \mathbb{R}^d$, $\mu_{t,x}$ is the (regular) conditional law of X_1 given $X_t = x$. In particular, if $\mathbb{E}[|X_1|] < \infty$,*

$$m(t, X_t) = \mathbb{E}[X_1 \mid X_t]. \quad (2.4)$$

Proof. The asserted claim follows once we show that

$$\mathbb{E}[g_1(X_1)g_2(X_t)] = \mathbb{E} \left[\int_{\mathbb{R}^d} g_1(y) \mu_{t,X_t}(dy) g_2(X_t) \right]$$

for any bounded Borel functions $g_i : \mathbb{R}^d \rightarrow \mathbb{R}$, $i = 1, 2$. By Proposition 2.1,

$$\mathbb{E}[g_1(X_1)g_2(X_t)] = \int_{\mathbb{R}^d \times \mathbb{R}^d} g_1(y) g_2(ty + \sqrt{t(1-t)}z) \phi_{I_d}(z) \mu(dy) dz$$

$$= \int_{\mathbb{R}^d \times \mathbb{R}^d} g_1(y) g_2(x) \phi_{t(1-t)I_d}(x - ty) \mu(dy) dx.$$

Meanwhile, observe that

$$\frac{d\mu_{t,x}}{d\mu}(y) = \frac{e^{-\frac{|ty-x|^2}{2t(1-t)}}}{\int_{\mathbb{R}^d} e^{-\frac{|tz-x|^2}{2t(1-t)}} \mu(dz)} = \frac{\phi_{t(1-t)I_d}(x - ty)}{p_t(x)}.$$

Hence

$$\begin{aligned} \mathbb{E} \left[\int_{\mathbb{R}^d} g_1(y) \mu_{t,X_t}(dy) g_2(X_t) \right] &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} g_1(y) \mu_{t,x}(dy) \right) p_t(x) g_2(x) dx \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} g_1(y) \phi_{t(1-t)I_d}(x - ty) \mu(dy) \right) g_2(x) dx. \end{aligned}$$

So the desired result follows. $\square$

When $\mathbb{E}[|X_1|] < \infty$, we have $\mathbb{E}[X_1 \mid \mathcal{F}_t^X] \rightarrow \mathbb{E}[X_1]$ a.s. as $t \downarrow 0$ by [51, Corollary II.2.4] and [35, Corollary 2.7.8]. Hence, $v_t \rightarrow \mathbb{E}[X_1]$ as $t \downarrow 0$ a.s. by (2.3), (2.4) and the Markov property of X . For this reason, we set $v_0 := \mathbb{E}[X_1]$ whenever $\mathbb{E}[|X_1|] < \infty$.

Remark 2.3 (Stochastic localization). The measure-valued stochastic process $(\mu_{t,X_t})_{t \in (0,1)}$ is closely related to Eldan [15]’s stochastic localization; see [37, Section 4] for details. Benton *et al.* [3] successfully utilize stochastic localization to obtain a refined error bound for discretization of the reverse SDE in SGM.

The following three propositions are useful consequences of (2.3) and Proposition 2.5.

Proposition 2.6 (Eldan *et al.* [19], Lemma 11). *Assume $\mathbb{E}[|X_1|^2] < \infty$. Then for every $t \in [0, 1)$,*

$$\mathbb{E}[|v_t|^2] = \frac{d - \mathbb{E}[\text{tr}(\Gamma_t)]}{1 - t} + \mathbb{E}[|X_1|^2] - d, \quad (2.5)$$

where

$$\Gamma_t := \frac{\text{Cov}[X_1 \mid X_t]}{1 - t}. \quad (2.6)$$

Proof. Using (2.3) and (2.4), we deduce

$$\mathbb{E}[|v_t|^2] = \frac{\mathbb{E}[|\mathbb{E}[X_1 \mid X_t] - X_t|^2]}{(1 - t)^2} = \frac{\mathbb{E}[|\mathbb{E}[X_1 \mid X_t]|^2] - 2\mathbb{E}[X_1 \cdot X_t] + \mathbb{E}[|X_t|^2]}{(1 - t)^2}.$$

By Proposition 2.1, we have $\mathbb{E}[X_1 \cdot X_t] = t \mathbb{E}[|X_1|^2]$ and $\mathbb{E}[|X_t|^2] = t^2 \mathbb{E}[|X_1|^2] + dt(1 - t)$. Consequently,

$$\mathbb{E}[|v_t|^2] = \frac{-\mathbb{E}[\text{tr}(\text{Cov}[X_1 \mid X_t])] + (1 - t)^2 \mathbb{E}[|X_1|^2] + dt(1 - t)}{(1 - t)^2} = \frac{d - \mathbb{E}[\text{tr}(\Gamma_t)]}{1 - t} + \mathbb{E}[|X_1|^2] - d.$$

$\square$

Proposition 2.7 (Eldan *et al.* [17], Lemma 13). *Assume $\mathbb{E}[|X_1|^2] < \infty$. Then, $\mathbb{E}[\int_0^1 \text{tr}(\Gamma_t^2) dt] < \infty$ and*

$$X_1 = \mathbb{E}[X_1] + \int_0^1 \Gamma_t dW_t^X.$$

Proof. Fix $\varepsilon \in (0, 1/2)$ arbitrarily. It is straightforward to verify that the map $[\varepsilon, 1 - \varepsilon] \times \mathbb{R}^d \ni (t, x) \mapsto m(t, x) \in \mathbb{R}^d$ is of class $C^{1,2}$. Hence, Ito's formula gives

$$m(t, X_t) = m_i(\varepsilon, X_\varepsilon) + \int_\varepsilon^t \frac{\partial m}{\partial s}(s, X_s) ds + \sum_{j=1}^d \int_\varepsilon^t \frac{\partial m_i}{\partial x_j}(s, X_s) dX_s^j + \frac{1}{2} \sum_{j=1}^d \int_\varepsilon^t \frac{\partial^2 m}{\partial x_j^2}(s, X_s) ds \quad (2.7)$$

for all $t \in [\varepsilon, 1 - \varepsilon]$, where X_s^j is the j -th component of X_s . Meanwhile, since $m(t, X_t) = \mathbb{E}[X_1 | \mathcal{F}_t^X]$ for any $t \in (0, 1)$ by (2.4) and the Markov property of X , $(m(t, X_t))_{t \in [\varepsilon, 1 - \varepsilon]}$ is a continuous martingale with respect to $(\mathcal{F}_t^X)_{t \in [\varepsilon, 1 - \varepsilon]}$. Hence, (2.7) and [51, Proposition IV.1.2] give

$$\mathbb{E}[X_1 | \mathcal{F}_t^X] = \mathbb{E}[X_1 | \mathcal{F}_\varepsilon^X] + \sum_{j=1}^d \int_\varepsilon^t \frac{\partial m}{\partial x_j}(s, X_s) dW_s^{X,j}, \quad (2.8)$$

where $W_s^{X,j}$ is the j -th component of W_s^X . A straightforward computation shows

$$\frac{\partial m_k}{\partial x_j}(s, x) = \frac{1}{1-s} \int_{\mathbb{R}^d} (y_j y_k - m_j(s, x) m_k(s, x)) \mu_{s,x}(dy),$$

where $m_k(s, x)$ denote the k -th component of $m(s, x)$. Combining this with (2.8) and Proposition 2.5 gives

$$\mathbb{E}[X_1 | \mathcal{F}_{1-\varepsilon}^X] = \mathbb{E}[X_1 | \mathcal{F}_\varepsilon^X] + \int_\varepsilon^{1-\varepsilon} \Gamma_t dW_t^X. \quad (2.9)$$

Using this expression, we obtain

$$\mathbb{E} \left[\int_\varepsilon^{1-\varepsilon} \text{tr}(\Gamma_t^2) dt \right] = \mathbb{E}[|\mathbb{E}[X_1 | \mathcal{F}_{1-\varepsilon}^X] - \mathbb{E}[X_1 | \mathcal{F}_\varepsilon^X]|^2] \leq \mathbb{E}[|X_1|^2],$$

where the first equality is by Proposition 1.23 in [51, Chapter IV] and the last inequality is by Jensen's inequality. Therefore, Fatou's lemma gives $\mathbb{E}[\int_0^1 \text{tr}(\Gamma_t^2) dt] \leq \mathbb{E}[|X_1|^2] < \infty$. This particularly implies that the stochastic integral $\int_0^1 \Gamma_t dW_t^X$ is well-defined. Then, since $\mathbb{E}[X_1 | \mathcal{F}_{1-\varepsilon}^X] \rightarrow X_1$ and $\mathbb{E}[X_1 | \mathcal{F}_\varepsilon^X] \rightarrow \mathbb{E}[X_1]$ a.s. as $\varepsilon \downarrow 0$ by [51, Corollary II.2.4] and [35, Corollary 2.7.8], we obtain the desired result by letting $\varepsilon \downarrow 0$ in (2.9). $\square$

Proposition 2.8 ([16], Fact 2.2). *Assume $\mathbb{E}[|X_1|] < \infty$. Then $(v_t)_{t \in [0,1]}$ is an $\mathbf{F}^X$ -martingale in $\mathbb{R}^d$.*

Proof. See Section A.3. $\square$

We conclude this section by a simple criterion for the finiteness of moments of the Föllmer drift.

Proposition 2.9. *For any $r > 0$, the following three conditions are equivalent: (i) $\mathbb{E}[|X_1|^r] < \infty$; (ii) $\mathbb{E}[|v_t|^r] < \infty$ for all $t \in (0, 1)$; (iii) $\mathbb{E}[|v_t|^r] < \infty$ for some $t \in (0, 1)$.*

Proof. See Section A.4. $\square$

3 Connection to the denoising diffusion probabilistic model

We begin by introducing the SDE formulation of the DDPM formally. Let $Y = (Y_t)_{t \geq 0}$ be a solution to the following SDE:

$$dY_t = -Y_t dt + \sqrt{2} dW_t, \quad Y_0 \sim \mu. \quad (3.1)$$

For $T > 0$, let $\bar{Y} = (\bar{Y}_t)_{t \in [0, T]}$ be the time reversal of $(Y_t)_{t \in [0, T]}$, i.e. $\bar{Y}_t = Y_{T-t}$ for $t \in [0, T]$. Then, by [28, Theorem 2.1], [51, Proposition VII.2.4] and Lévy's characterization, there exists a standard Wiener process $\bar{W} = (\bar{W}_t)_{t \in [0, T]}$ in $\mathbb{R}^d$ such that

$$\bar{Y}_t = \bar{Y}_0 + \int_0^t (\bar{Y}_s + 2\nabla \log q_{T-s}(\bar{Y}_s)) ds + \sqrt{2} \bar{W}_t \quad (3.2)$$

for all $t \in [0, T)$, where q_t is the density of Y_t for every $t > 0$.

As already mentioned in the introduction, the Föllmer process can be seen as an “augmented” time-compressed version of the reverse SDE (3.2). This fact is essentially known in the literature, as the Föllmer process is an (augmented) time-compressed version of Eldan's stochastic localization process (see [37, Section 4.2]) and the process $\bar{Y}$ in (3.2) is a change of variables of the latter (see [47, Section 1.3] or [53, Theorem 3]). The following proposition formulates this fact more explicitly:

Proposition 3.1. *For $T > 0$, define a process $\bar{Y} = (\bar{Y}_t)_{t \in [0, T]}$ as $\bar{Y}_t = e^{T-t} X_{e^{2(t-T)}}$, $t \in [0, T]$. Then $\bar{Y}_0 \sim \mathcal{S}_{e^{-2T}} \mu$ and the process*

$$\bar{W}_t = \frac{1}{\sqrt{2}} \left(\bar{Y}_t - \bar{Y}_0 - \int_0^t (\bar{Y}_s + 2\nabla \log q_{T-s}(\bar{Y}_s)) ds \right) \quad (t \in [0, T])$$

is a standard $\mathbb{F}^{\bar{Y}}$ -Wiener process in $\mathbb{R}^d$.

Proof. The first claim is immediate from the definition of $\bar{Y}$. To prove the second claim, define a process $V = (V_t)_{t \geq 0}$ as $V_t = W_{e^{2(t-T)}}^X$ for $t \in [0, T]$. By Proposition V.1.5 in [51], V is an $(\mathcal{F}_{e^{2(t-T)}}^X)$ -martingale in $\mathbb{R}^d$ and $[V, V]_t = e^{2(t-T)} I_d$ for every $t \in [0, T]$. Hence the process

$$\widetilde{W}_t = \frac{1}{\sqrt{2}} \int_0^t e^{T-s} dV_s \quad (t \in [0, T])$$

is an $(\mathcal{F}_{e^{2(t-T)}}^X)$ -Wiener process in $\mathbb{R}^d$ by Lévy's characterization. Since $\mathcal{F}_{e^{2(t-T)}}^X = \mathcal{F}_t^{\bar{Y}}$ for all $t \in [0, T]$, we complete the proof once we show that $\bar{W}_t = \widetilde{W}_t$ for every $t \in [0, T]$.

Observe that $q_t(y) = p_{e^{-2t}}(e^{-t}y)$ for $t > 0$ and $y \in \mathbb{R}^d$. Hence

$$\begin{aligned} \int_{e^{-2T}}^{e^{2(t-T)}} \left(\frac{X_s}{s} + \nabla \log p_s(X_s) \right) ds &= 2 \int_0^t \left(\frac{X_{e^{2(u-T)}}}{e^{2(u-T)}} + \nabla \log p_{e^{2(u-T)}}(X_{e^{2(u-T)}}) \right) e^{2(u-T)} du \\ &= 2 \int_0^t (\bar{Y}_u + \nabla \log q_{T-u}(\bar{Y}_u)) e^{u-T} du. \end{aligned}$$

Therefore, integration by parts gives

$$\bar{Y}_t - \bar{Y}_0 = 2 \int_0^t (\bar{Y}_u + \nabla \log q_{T-u}(\bar{Y}_u)) du + \sqrt{2} \widetilde{W}_t - \int_0^t \bar{Y}_u du$$

$$= \int_0^t (\bar{Y}_u + 2\nabla \log q_{T-u}(\bar{Y}_u)) du + \sqrt{2}\widetilde{W}_t.$$

Hence $\bar{W}_t = \widetilde{W}_t$. $\square$

Proposition 3.1 suggests that discretization of the Föllmer process could be used to analyze sampling errors of the DDPM. In this section, we implement this analysis while surveying relevant error bounds for the DDPM sampler.

3.1 Discretized Föllmer process and the DDPM sampler

Given i.i.d. data generated from μ , we aim to construct a generator that simulates a random vector whose distribution is close to μ . Suppose that we have a good estimator $\widehat{\mathfrak{s}}(x, t)$ for $\nabla \log p_t(x)$. Then, by simulating a solution $\widehat{X} = (\widehat{X}_t)_{t \in [0,1]}$ to the SDE

$$d\widehat{X}_t = \left(\frac{\widehat{X}_t}{t} + \widehat{\mathfrak{s}}(\widehat{X}_t, t) \right) dt + dW_t \quad (0 \leq t \leq 1), \quad \widehat{X}_0 = 0, \quad (3.3)$$

we expect that the law of $\widehat{X}_1$ would be close to μ . In practice, we need to discretize the SDE (3.3) for the simulation. Below we discuss error bounds for such discretization.

Let $0 < t_0 < t_1 < \dots < t_N \leq 1$ be time points for discretization and set $\delta := 1 - t_N$. We will perform early stopping so that $\delta > 0$ when μ is not smooth enough because the score $\nabla \log p_t(x)$ may blow up as $t \uparrow 1$ in such a situation. Also, analogously to [3], we introduce the following conditions:

[A1] There exists a constant $\kappa > 0$ such that

$$h_i := t_{i+1} - t_i \leq \kappa(1 - t_{i+1}) \quad (i = 0, 1, \dots, N). \quad (3.4)$$

[A2] The score estimator $\widehat{\mathfrak{s}} : \mathbb{R}^d \times (0, 1) \rightarrow \mathbb{R}^d$ is measurable and satisfies

$$\sum_{i=0}^{N-1} (t_{i+1} - t_i) \mathbb{E} \left[|\nabla \log p_{t_i}(X_{t_i}) - \widehat{\mathfrak{s}}(X_{t_i}, t_i)|^2 \right] \leq \varepsilon_{\text{score}}^2 \quad (3.5)$$

for some $\varepsilon_{\text{score}} \geq 0$.

We remark that (3.4) and (3.5) are implied by the corresponding conditions in terms of the process $\bar{Y}_t = e^{T-t} X_{e^{2(t-T)}}$, $t \in [0, T]$, as shown in the following lemma. Observe that the sampling times corresponding to $\bar{Y}$ are given by $\tau_i = T + \frac{1}{2} \log t_i$ for $i = 0, 1, \dots, N$.

Lemma 3.1. *Assume*

$$\tau_{i+1} - \tau_i \leq \kappa \min\{1, T - \tau_{i+1}\} \quad (i = 0, 1, \dots, N-1) \quad (3.6)$$

for some constant $\kappa > 0$. Then we have (3.4). Further, suppose that a measurable function $\widetilde{\mathfrak{s}} : \mathbb{R}^d \times (0, T) \rightarrow \mathbb{R}^d$ satisfies

$$\sum_{i=0}^{N-1} (\tau_{i+1} - \tau_i) \mathbb{E} \left[|\nabla \log q_{T-\tau_i}(\bar{Y}_{\tau_i}) - \widetilde{\mathfrak{s}}(\bar{Y}_{\tau_i}, T - \tau_i)|^2 \right] \leq \varepsilon^2 \quad (3.7)$$

for some $\varepsilon \geq 0$. Then we have (3.5) with $\widehat{\mathfrak{s}}(x, t) = \frac{1}{\sqrt{t}} \widetilde{\mathfrak{s}}\left(\frac{x}{\sqrt{t}}, -\frac{1}{2} \log t\right)$ and $\varepsilon_{\text{score}} = \sqrt{2}e^\kappa \varepsilon$.

Proof. Observe that

$$\tau_{i+1} - \tau_i = \frac{1}{2} \log(t_{i+1}/t_i) \quad \text{and} \quad T - \tau_{i+1} = \frac{1}{2} \log(1/t_{i+1}).$$

Hence,

$$t_{i+1} - t_i = t_{i+1}(1 - t_i/t_{i+1}) \leq 2t_{i+1}(\tau_{i+1} - \tau_i) \leq \kappa t_{i+1} \log(1/t_{i+1}) \leq \kappa(1 - t_{i+1}),$$

where the first and third inequalities follow by $\log x \leq x - 1$ for any $x > 0$ and the second by (3.6). Thus, we have (3.4). Also, since

$$\begin{aligned} & \sum_{i=0}^{N-1} (t_{i+1} - t_i) \mathbb{E} \left[|\nabla \log p_{t_i}(X_{t_i}) - \widehat{\mathbf{s}}(X_{t_i}, t_i)|^2 \right] \\ &= \sum_{i=0}^{N-1} (1 - t_i/t_{i+1}) \frac{t_{i+1}}{t_i} \mathbb{E} \left[|\nabla \log q_{T-\tau_i}(Y_{T-\tau_i}) - \widetilde{\mathbf{s}}(Y_{T-\tau_i}, T - \tau_i)|^2 \right] \\ &\leq 2e^{2\kappa} \sum_{i=0}^{N-1} (\tau_{i+1} - \tau_i) \mathbb{E} \left[|\nabla \log q_{T-\tau_i}(Y_{T-\tau_i}) - \widetilde{\mathbf{s}}(Y_{T-\tau_i}, T - \tau_i)|^2 \right], \end{aligned}$$

we have (3.5) with $\varepsilon_{\text{score}} = \sqrt{2}e^{\kappa}\varepsilon$. $\square$

We also note that $T - \tau_N$ has the same order as δ . In fact, the inequality $1 - x \leq \log(1/x) \leq (1 - x)/x$ gives

$$\frac{\delta}{2} \leq T - \tau_N \leq \frac{\delta}{2(1 - \delta)}.$$

Remark 3.1 (Score estimation). The score estimator $\widehat{\mathbf{s}}$ is typically constructed by minimizing the left hand side of (3.5) (or (3.7)) with respect to $\widehat{\mathbf{s}}$ over a class of neural networks. It is known that for all $0 < t < 1$,

$$\begin{aligned} & \mathbb{E} \left[|\nabla \log p_t(X_t) - \widehat{\mathbf{s}}(X_t, t)|^2 \right] \\ &= \mathbb{E} \left[\left| \frac{-Z}{\sqrt{t(1-t)}} - \widehat{\mathbf{s}}(\xi_t, t) \right|^2 \right] - \mathbb{E} \left[\left| \frac{-Z}{\sqrt{t(1-t)}} - \nabla \log p_t(\xi_t) \right|^2 \right], \end{aligned} \quad (3.8)$$

where $\xi_t = t\xi + \sqrt{t(1-t)}Z$ with $\xi \sim \mu$ and $Z \sim \gamma_d$ being independent; see Section A.5 for a proof. Since the second term of the right hand side on (3.8) is independent of $\widehat{\mathbf{s}}$, the above minimization problem is equivalent to minimizing

$$\sum_{i=0}^{N-1} (t_{i+1} - t_i) \mathbb{E} \left[\left| \frac{-Z}{\sqrt{t_i(1-t_i)}} - \widehat{\mathbf{s}}(\xi_{t_i}, t_i) \right|^2 \right]$$

with respect to $\widehat{\mathbf{s}}$. Further, since μ is unknown, one usually replaces μ by the empirical distribution of observation data. The estimation error induced by this procedure can be studied separately and we do not pursue this topic here. We refer to [27, 48] for recent developments of this topic.

To discretize the SDE (3.3), we first focus on the Euler–Maruyama scheme. Namely, we construct a discretized process $(X_{t_i}^{\text{EM}})_{i=0}^N$ as

$$\begin{cases} X_{t_0}^{\text{EM}} = \sqrt{t_0} Z_0, \\ X_{t_{i+1}}^{\text{EM}} = X_{t_i}^{\text{EM}} + \left(\frac{X_{t_i}^{\text{EM}}}{t_i} + \widehat{\mathbf{s}}(X_{t_i}^{\text{EM}}, t_i) \right) h_i + \sqrt{h_i} Z_i \quad (i = 0, 1, \dots, N-1), \end{cases}$$

where $Z_i \stackrel{i.i.d.}{\sim} N(0, I_d)$. As anticipated from Proposition 3.1, this algorithm can be interpreted as a special case of the DDPM sampler. To formulate this precisely, we introduce the DDPM sampler in a general form, following [40]. Given constants $\alpha_i \in (0, 1)$ ($i = 0, 1, \dots, N$), suppose that we have an estimator $\mathbf{s}_i(x)$ for the score function of $\mathcal{S}_{\bar{\alpha}_i} \mu$ for each $i = 0, 1, \dots, N-1$, where $\bar{\alpha}_i := \prod_{j=i}^N \alpha_j$. Then, the DDPM sampler sequentially generates random vectors $\mathbf{x}_i$ ($i = 0, 1, \dots, N$) according to the following update rule:

$$\mathbf{x}_0 \sim N(0, I_d), \quad \mathbf{x}_{i+1} = \frac{1}{\sqrt{\alpha_i}} (\mathbf{x}_i + \eta_i \mathbf{s}_i(\mathbf{x}_i) + \sigma_i Z_i) \quad (i = 0, 1, \dots, N-1), \quad (3.9)$$

where $\eta_i, \sigma_i > 0$ are hyper-parameters chosen so that the law of $\mathbf{x}_i$ is close to $\mathcal{S}_{\bar{\alpha}_i} \mu$ for each i . Noting that $X_t/\sqrt{t} \sim \mathcal{S}_t \mu$ for each $t \in (0, 1]$, one can easily verify that $\mathbf{x}_i = X_{t_i}^{\text{EM}}/\sqrt{t_i}$ ($i = 0, 1, \dots, N$) correspond to samples from the DDPM sampler with $\mathbf{s}_i(x) = \sqrt{t_i} \widehat{\mathbf{s}}(\sqrt{t_i} x, t_i)$, $\alpha_i = t_i/t_{i+1}$, $\eta_i = 1 - \alpha_i$, $\sigma_i = \sqrt{\alpha_i(1 - \alpha_i)}$ for $i = 0, 1, \dots, N-1$ and $\alpha_N = t_N$. Interestingly, this hyper-parameter setting is the *exactly* same as the standard choice used in the discrete-time model, originally proposed in [29] (see also Section 2.2 of [54]). By contrast, the *exponential integrator scheme*, a popular scheme to discretize the SDE (3.2), leads to the following hyper-parameter setting: $\alpha_i = e^{-2(\tau_{i+1} - \tau_i)} = t_i/t_{i+1}$, $\eta_i = 2(1 - \sqrt{\alpha_i})$, $\sigma_i = \sqrt{1 - \alpha_i}$ for $i = 0, 1, \dots, N-1$ and $\alpha_N = e^{-2(T - \tau_N)} = t_N$. See e.g. Eq.(5) in [11] for the explicit construction of the exponential integrator scheme. This is only approximately equivalent to the standard choice above.

We proceed to giving concrete bounds for discretization errors. The first result is a counterpart of Theorem 2 in Benton *et al.* [3]:

Theorem 3.1. *Assume [A1]–[A2] and $t_0 \leq 1/2$. Then*

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{EM}}}) \leq \varepsilon_{\text{score}}^2 + \kappa d \log \left(\frac{1 - t_0}{\delta} \right) + 2\kappa \mathbb{E}[|X_1|^2] + \frac{t_0}{2} (d + \mathbb{E}[|X_1|^2]).$$

The proof of Theorem 3.1 is given in Section A.6. We note that the bound in Theorem 3.1 is applicable to $\mathbf{x}_N = X_{t_N}^{\text{EM}}/\sqrt{t_N}$ because $H(\mathcal{S}_{t_N} \mu \mid P^{\mathbf{x}_N}) = H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{EM}}})$ by Lemma A.4.

Remark 3.2 (Comparison to Benton *et al.* [3]). When $\mathbf{x}_i$ are obtained via discretizing $\bar{Y}_t = e^{T-t} X_{e^{2(t-T)}}$ by the exponential integrator scheme with sampling times $\tau_i = T + \frac{1}{2} \log t_i$, Theorem 2 in [3] gives the following upper bound for $H(\mathcal{S}_{t_N} \mu \mid P^{\mathbf{x}_N})$ up to a universal constant under (3.6) and (3.7):

$$\varepsilon^2 + \kappa^2 d N + \kappa d T + \kappa \mathbb{E}[|X_1|^2] + (d + \mathbb{E}[|X_1|^2]) e^{-2T}. \quad (3.10)$$

Recall that T is related to t_0 as $t_0 = e^{-2T}$. Also, since

$$(\kappa N + 1)\delta \geq \sum_{i=0}^{N-1} (t_{i+1} - t_i) + 1 - t_N = 1 - t_0,$$

we have $\log((1 - t_0)/\delta) \leq (1 - t_0)/\delta - 1 \leq \kappa N$. Therefore, thanks to Lemma 3.1, Theorem 3.1 gives the following bound for $H(\mathcal{S}_{t_N}\mu \mid P^{\times N})$ up to a universal constant under the same assumption as above with assuming $\kappa \leq 1$:

$$\varepsilon^2 + \kappa^2 dN + \kappa \mathbb{E}[|X_1|^2] + (d + \mathbb{E}[|X_1|^2]) e^{-2T}.$$

Compared to (3.10), our bound does not contain the term κdT . In particular, except for implicit dependence of the score estimation error on T (see Remark 3.1), a large value of T does not hurt our bound. In view of the correspondence to the DDPM sampler discussed above, this is natural because the parameters in (3.9) is determined via ratios t_i/t_{i+1} which do not depend on T . Note that the appearance of e^{-2T} in the bounds is due to the initialization error.

Remark 3.3. Recently, Jiao *et al.* [34] have established a bound for $H(\mathcal{S}_{t_N}\mu \mid P^{\times N})$ of order $O((\kappa d)^2)$ with respect to κ from (3.6) and d up to some logarithmic factor (see also [32]). Their proof approach is fundamentally different from ours: They do not view the DDPM sampler as a discretized SDE but construct the corresponding “true” denoising process via a noise-injected probability flow ordinary differential equation of [54] (or the Kim–Milman map in terms of [37]). It is unclear whether our approach can give a similar bound.

When μ has a finite relative Fisher information with respect to γ_d , we can bound the discretization error without early stopping. The next result is a counterpart of Theorem 2.1 in Conforti *et al.* [8]:

Theorem 3.2. Assume [A2], $t_0 \leq 1/2$, $t_N = 1$ and that μ has an almost everywhere differentiable density. Then, with $\bar{h} := \max_{i=0,1,\dots,N-1} h_i$,

$$H(\mu \mid P^{X_1^{\text{EM}}}) \leq t_0 H(\mu \mid \gamma_d) + \varepsilon_{\text{score}}^2 + I(\mu \mid \gamma_d) \bar{h}.$$

We show this result in Section A.7.

Remark 3.4 (Comparison to Conforti *et al.* [8]). When x_i are obtained as in Remark 3.2 and τ_i are equidistant, Theorem 2.1 in [8] gives the following upper bound for $H(\mathcal{S}_{t_N}\mu \mid P^{\times N})$ up to a universal constant under (3.7):

$$e^{-2T} H(\mu \mid \gamma_d) + \varepsilon^2 + I(\mu \mid \gamma_d) \frac{T}{N}.$$

Since the assumptions of Theorem 3.2 are satisfied with $\bar{h} = T/N$ in this case, our result recovers this bound. In addition, if we take t_i to be equidistant instead of τ_i , the assumptions of Theorem 3.2 are satisfied with $\bar{h} = 1/(N+1)$, so the coefficient of $I(\mu \mid \gamma_d)$ is improved while keeping the number of sampling steps N .

3.2 Adaptation to intrinsic dimensionality

In practice, data are often distributed around a low-dimensional manifold. Recently, Li & Yan [40] have shown that the DDPM sampler (3.9) must satisfy

$$\eta_i = 1 - \alpha_i, \quad \sigma_i^2 = \frac{(1 - \alpha_i)(\alpha_i - \bar{\alpha}_i)}{1 - \bar{\alpha}_i} \quad (i = 0, 1, \dots, N-1) \quad (3.11)$$

for $H(P^{\times N} \mid \mathcal{S}_{\alpha_N} \mu)$ having a bound independent of the ambient dimension d ; see Theorem 2 *ibidem*.² Azangulov *et al.* [1] and Huang *et al.* [31] give an interpretation for this specification from the SDE perspective. In particular, Huang *et al.* [31] have shown that this specification is interpreted as discretizing only the posterior mean component in the drift of (3.2). For the Föllmer process, this corresponds to discretizing only $m(t, X_t)$ in the expression (2.3). This leads to the following linear SDE:

$$\begin{cases} \tilde{X}_0 = 0, \\ d\tilde{X}_t = \frac{m(t_i, \tilde{X}_{t_i}) - \tilde{X}_t}{1-t} dt + dW_t \quad \text{when } t \in [t_i, t_{i+1}) \text{ for some } i = -1, 0, \dots, N-1, \end{cases} \quad (3.12)$$

where we set $t_{-1} := 0$ and $m(0, x) \equiv 0$ by convention. Note that although (2.4) suggests setting $m(0, x) \equiv \mathbb{E}[X_1]$ when $\mathbb{E}[|X_1|] < \infty$, we always set $m(0, x) \equiv 0$ because $\mathbb{E}[X_1]$ is unknown in practice. The above SDE can be explicitly solved as

$$\tilde{X}_t = \frac{1-t}{1-t_i} \left(\tilde{X}_{t_i} + \frac{t-t_i}{1-t} m(t_i, \tilde{X}_{t_i}) + \int_{t_i}^t \frac{1-t_i}{1-s} dW_s \right) \quad \text{for } t \in [t_i, t_{i+1}). \quad (3.13)$$

See e.g. Section 5.6 in [35]. Observe that

$$\frac{1-t_{i+1}}{1-t_i} \int_{t_i}^t \frac{1-t_i}{1-s} dW_s \sim N \left(0, (t_{i+1} - t_i) \frac{1-t_{i+1}}{1-t_i} I_d \right) \quad (i = -1, \dots, N-1)$$

and they are independent. Hence the stochastic integral term of (3.13) can be simulated exactly. Meanwhile, by (2.3) and the definition of v_t ,

$$m(t, x) = \frac{x}{t} + (1-t) \nabla \log p_t(x) \quad \text{for all } 0 < t < 1, x \in \mathbb{R}^d.$$

Replacing the score function by its estimate, we obtain the following alternative discretization scheme:

$$\begin{cases} X_{t_0}^{\text{ada}} = \sqrt{t_0(1-t_0)} Z_0, \\ X_{t_{i+1}}^{\text{ada}} = \frac{t_{i+1}}{t_i} X_{t_i}^{\text{ada}} + h_i \hat{\mathbf{s}}(X_{t_i}^{\text{ada}}, t_i) + \sqrt{h_i \frac{1-t_{i+1}}{1-t_i}} Z_i \quad (i = 0, 1, \dots, N-1). \end{cases}$$

One can easily check that if we change the initialization in (3.9) to $x_0 \sim N(0, \sqrt{1-t_0} I_d)$, $x_i = X_{t_i}^{\text{ada}} / \sqrt{t_i}$ ($i = 1, \dots, N$) satisfy (3.9) with $\mathbf{s}_i(x) = \sqrt{t_i} \hat{\mathbf{s}}(\sqrt{t_i} x, t_i)$, $\alpha_i = t_i / t_{i+1}$ for $i = 0, 1, \dots, N-1$, $\alpha_N = t_N$ and specification (3.11). Below we will see that the different initialization serves as avoiding the appearance of the ambient dimension d caused by initialization.

The following result gives a bound corresponding to Theorem 3.1 for $(X_{t_i}^{\text{ada}})_{i=0}^N$. We prove it in Section A.8.

Theorem 3.3. *Assume [A1]–[A2] and $t_0 \leq 1/2$. Then*

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + \kappa d \log \left(\frac{1-t_0}{\delta} \right) + 3\kappa \mathbb{E}[|X_1|^2] + \varepsilon_{\text{score}}^2.$$

²Interestingly, this choice matches the hyper-parameter setting optimized for the case that μ is a unit mass; see [29, Section 3.2].

Next we state a bound adaptive to a low-dimensionality structure of data as in [31]. We need to introduce some notation. Given a subset $A \subset \mathbb{R}^d$ and $\varepsilon > 0$, we call a subset $\mathcal{N} \subset A$ an ε -net of A if $A \subset \bigcup_{x \in \mathcal{N}} \mathcal{B}(x; \varepsilon)$, where $\mathcal{B}(x; \varepsilon) := \{y \in \mathbb{R}^d : |y - x| < \varepsilon\}$. The maximal cardinality of ε -nets of A is called the *covering number* of A and denoted by $\mathcal{N}(A, \varepsilon)$. Also, let $\mathcal{X}$ be the support of μ .

[A3] There exist constants $0 < \varepsilon_0 \leq e^{-1}$ and $k \geq 1$ such that

$$\mathcal{N}(\mathcal{X}, \varepsilon_0) \leq \varepsilon_0^{-k}, \quad \varepsilon_0 \leq \sqrt{\delta} \log(1/\varepsilon_0)$$

and

$$\mathbb{E} \left[\sup_{x, y \in \mathcal{X} : |x - y| \leq \varepsilon_0} |(x - y) \cdot Z| \right] \leq \sqrt{\delta} k \log(1/\varepsilon_0), \quad (3.14)$$

where $Z \sim N(0, 1)$.

[A4] $\mathbb{E}[|X_1|^4] < \infty$ and there exists a constant $R \geq 3$ such that $\mathbb{E}[|X_1 - \mathbb{E}[X_1]|^4] \leq R^4$.

Remark 3.5. (a) [A3] is essentially equivalent to Assumption 1 in [31]: A seemingly additional condition (3.14) is satisfied as long as ε_0 is sufficiently small because we always have

$$\mathbb{E} \left[\sup_{x, y \in \mathcal{X} : |x - y| \leq \varepsilon_0} |(x - y) \cdot Z| \right] \leq \varepsilon_0 \mathbb{E}[|Z|] \leq \varepsilon_0 \sqrt{d}. \quad (3.15)$$

We keep the condition (3.14) in this form because the left hand side of (3.14) is known as the *Gaussian width* of the set $\{x - y : x, y \in \mathcal{X}, |x - y| \leq \varepsilon_0\}$ whose bound is well-studied in the literature; see e.g. [58, Chapters 7–8]. See also Corollary 3.2.

(b) [A3] is closely related to the notion of *Minkowski dimension* (see e.g. [42, Chapter 5] for a detailed account). For a subset $A \subset \mathbb{R}^d$, the *upper Minkowski dimension* of A is defined as

$$\overline{\dim}_M(A) := \limsup_{\varepsilon \downarrow 0} \frac{\log \mathcal{N}(A, \varepsilon)}{\log(1/\varepsilon)}.$$

Also, for $\varepsilon > 0$, the ε -enlargement of A is defined as the set $\{y \in \mathbb{R}^d : |y - x| < \varepsilon \text{ for some } x \in A\}$. One can easily check that [A3] is satisfied if $\mathcal{X}$ is contained in the $(\varepsilon_0/2)$ -enlargement of a set $A \subset \mathbb{R}^d$ with $\overline{\dim}_M(A) < k/(1 + \log 2)$ for sufficiently small $\varepsilon_0 > 0$ (see also [31, Lemma 3]). We have $\overline{\dim}_M(A) \leq m$ for some $m \in \mathbb{N}$ when $A = g(B)$ for some bounded $B \subset \mathbb{R}^m$ and Lipschitz map $g : B \rightarrow \mathbb{R}^n$. This directly follows from definition once we note that $\overline{\dim}_M(B) \leq m$ for any bounded subset $B \subset \mathbb{R}^m$ by Eq.(4.17) in [58]. More precisely, if the diameter of B is bounded by $D \geq 1$ and the Lipschitz constant of g is bounded by $L \geq 1$, then $\mathcal{N}(A, \varepsilon) \leq \varepsilon^{-m - \log(3DL)}$ for all $0 < \varepsilon < e^{-1}$. Huang *et al.* [31] give more examples satisfying [A3].

Theorem 3.4. Assume [A1]–[A4] and $t_0 \leq 1/2$. There exists a universal constant $C > 0$ such that

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + 4\kappa \mathbb{E}[|X_1|^2] + C\kappa L \log \left(\frac{1 - t_0}{\delta} \right) + \varepsilon_{\text{score}}^2, \quad (3.16)$$

where $L := k \log(1/\varepsilon_0) + \log(R/\delta)$. If we further assume

$$h_i \leq 6\kappa t_i \quad (i = 0, 1, \dots, N-1), \quad (3.17)$$

then $4\kappa \mathbb{E}[|X_1|^2]$ in (3.16) can be replaced by $C\kappa L \log(1/t_0)$.

This result is proven in Section A.9. Observe that $h_i/t_i \leq 2e^{\tau_{i+1}-\tau_i}(\tau_{i+1}-\tau_i)$ for every i . Hence (3.17) follows from (3.6) if $\kappa \leq 1$. To compare our bound to [31]’s one, we focus on the case that the order of R, δ, d and ε_0 are controlled by k . Noting Remark 3.2 and (3.15), we have the following corollary:

Corollary 3.1. *Assume [A1]–[A4], $t_0 \leq 1/2$ and (3.17). Moreover, suppose that there exists constants $c_1, c_2 \geq 1$ such that $R \leq k^{c_1 k}, \delta \geq k^{-c_2}, d \leq k^{c_2}$ and $\varepsilon_0 = k^{1-c_2}$. Then we have*

$$H(P^{X_{t_N}} | P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + C_1 \kappa \min \{ \mathbb{E}[|X_1|^2], (k \log k) \log(1/t_0) \} + C_1 \kappa^2 N k \log k + \varepsilon_{\text{score}}^2,$$

where $C_1 > 0$ is a constant depending only on c_1 and c_2 .

Remark 3.6 (Comparison to Huang *et al.* [31]). Under similar assumptions to Corollary 3.1, Theorem 2 in [31] gives the same bound as in Corollary 3.1 but with an additional term $t_0 d$ up to a constant. This improvement comes from the difference in their initialization steps. In fact, the standard initialization corresponds to using the Euler–Maruyama scheme to discretize the “augmented” process $(X_t)_{t \in [0, t_0]}$ in the Föllmer process, causing the $t_0 (d + \mathbb{E}[|X_1|^2])$ error. We also remark that our assumption on R is weaker than that of [31]. In fact, the latter assumes that R is an upper bound of the diameter of $\mathcal{X}$ and of polynomial growth in k .

The following corollary states the bound in terms of the Minkowski dimension of $\mathcal{X}$ and completely avoids any growth conditions on d .

Corollary 3.2. *Assume [A1]–[A2], [A4] and $t_0 \leq 1/2$. Also, assume that there exist constants $k, c \geq 1$ such that $\mathcal{N}(\mathcal{X}, \varepsilon) \leq c\varepsilon^{-k}$ for all $0 < \varepsilon \leq e^{-1}$ and $R \leq c\delta^{-k}$. Then we have*

$$H(P^{X_{t_N}} | P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + 4\kappa \mathbb{E}[|X_1|^2] + C_2 \kappa k \log(1/\delta) \log\left(\frac{1-t_0}{\delta}\right) + \varepsilon_{\text{score}}^2, \quad (3.18)$$

where C_2 is a constant depending only on c . If we further assume (3.17), then $4\kappa \mathbb{E}[|X_1|^2]$ in (3.18) can be replaced by $C_2 \kappa k \log(1/\delta) \log(1/t_0)$.

Proof. By Dudley’s maximal inequality (e.g. Theorem 2.2.4 in [56]), there exists a universal constant $C > 0$ such that

$$\mathbb{E} \left[\sup_{x, y \in \mathcal{X}: |x-y| \leq \varepsilon_0} |(x-y) \cdot Z| \right] \leq C \int_0^{\varepsilon_0} \sqrt{\log \mathcal{N}(\mathcal{X}, \varepsilon/2)} d\varepsilon$$

for any $\varepsilon_0 > 0$. Therefore, for any $0 < \varepsilon_0 \leq e^{-1}$,

$$\mathbb{E} \left[\sup_{x, y \in \mathcal{X}: |x-y| \leq \varepsilon_0} |(x-y) \cdot Z| \right] \leq C \left(\varepsilon_0 \sqrt{\log c} + \int_0^{\varepsilon_0/2} \sqrt{k \log(1/\varepsilon)} d\varepsilon \right) \leq C' \sqrt{k} \varepsilon_0 \sqrt{\log(1/\varepsilon_0)},$$

where $C' \geq 1$ is a constant depending only on c . Hence [A3] is satisfied with $\varepsilon_0 = \min\{\sqrt{\delta}/C', e^{-1}\}$ and k replaced by $k + \log c$. Consequently, the desired result follows from Theorem 3.4. $\square$

A Additional proofs

A.1 Proof of Theorem 2.1

We need two auxiliary results. The first one is the so-called *Tweedie's formula* [13, 14] that expresses scores as conditional expectations.

Lemma A.1 (Tweedie's formula). *Let Y and Z be independent random vectors in $\mathbb{R}^d$ such that $Z \sim N(0, \Sigma)$ for some positive definite symmetric matrix Σ . Then $Y + Z$ has a positive, smooth density f_{Y+Z} and*

$$\nabla \log f_{Y+Z}(Y + Z) = -\mathbb{E}[\Sigma^{-1}Z \mid Y + Z].$$

Proof. It is well-known that $f_{Y+Z}(x) = \mathbb{E}[\phi_\Sigma(x - Y)]$ is the density of $Y + Z$. So the first claim follows. We also obtain

$$\mathbb{E}[|\nabla \log f_{Y+Z}(Y + Z)|] = \int_{\mathbb{R}^d} |\nabla f_{Y+Z}(x)| dx \leq \mathbb{E} \left[\int_{\mathbb{R}^d} |\nabla \phi_\Sigma(x - Y)| dx \right] = \int_{\mathbb{R}^d} |\nabla \phi_\Sigma(x)| dx < \infty.$$

Hence the second claim follows once we show that

$$\mathbb{E}[\nabla \log f_{Y+Z}(Y + Z)g(Y + Z)] = -\mathbb{E}[\Sigma^{-1}Zg(Y + Z)]$$

for any compactly supported smooth function $g : \mathbb{R}^d \rightarrow \mathbb{R}$. Integration by parts gives

$$\mathbb{E}[\nabla \log f_{Y+Z}(Y + Z)g(Y + Z)] = \int_{\mathbb{R}^d} \nabla f_{Y+Z}(x)g(x)dx = - \int_{\mathbb{R}^d} f_{Y+Z}(x)\nabla g(x)dx.$$

Meanwhile, by the multivariate Stein identity

$$\mathbb{E}[\Sigma^{-1}Zg(Y + Z)] = \mathbb{E}[\nabla g(Y + Z)] = \int_{\mathbb{R}^d} \nabla g(x)f_{Y+Z}(x)dx.$$

Hence we obtain the desired result. $\square$

The second one is a consequence of the Markov property of X :

Lemma A.2. $\mathbb{E}[\beta_t \mid \mathcal{F}_t^X] = \mathbb{E}[\beta_t \mid X_t]$ for all $t \in [0, 1]$.

Proof. Define $\eta_n := \beta_t 1_{\{|\beta_t| \leq n\}}$ for each $n \in \mathbb{N}$. Since X is a Markov process and η_n is bounded and measurable with respect to $\sigma(X_s, s \in [t, 1])$, we have $\mathbb{E}[\eta_n \mid \mathcal{F}_t^X] = \mathbb{E}[\eta_n \mid X_t]$ by property (ia) in [7, Section 1.1]. Since $\eta_n \rightarrow \beta_t$ as $n \rightarrow \infty$, $|\eta_n| \leq |\beta_t|$ and $\mathbb{E}[|\beta_t|] < \infty$, we obtain the desired result by the dominated convergence theorem. $\square$

Proof of Theorem 2.1. Observe that

$$\mathbb{E} \left[\int_0^1 \frac{|\beta_t|}{1-t} dt \right] \leq \sqrt{d} \int_0^1 \frac{\sqrt{t}}{\sqrt{1-t}} dt < \infty.$$

Hence we can define a process $B = (B_t)_{t \in [0,1]}$ as

$$B_t = \beta_t + \int_0^t \frac{\beta_s}{1-s} ds \quad (0 \leq t \leq 1). \quad (\text{A.1})$$

Recall that β is a unique strong solution to the following SDE:

$$d\beta_t = \frac{-\beta_t}{1-t}dt + dW_t \quad (0 \leq t < 1), \quad \beta_0 = 0.$$

This fact and the continuity of B at $t = 1$ imply that B is a standard Wiener process in $\mathbb{R}^d$ such that $\mathbf{F}^B = \mathbf{F}^\beta$.

Now, by Tweedie's formula and Lemma A.2,

$$\nabla \log p_s(X_s) = -\frac{\mathbb{E}[\beta_s | X_s]}{s(1-s)} = -\frac{\mathbb{E}[\beta_s | \mathcal{F}_s^X]}{s(1-s)}. \quad (\text{A.2})$$

Hence

$$W_t^X = tX_1 + \beta_t - \int_0^t \left(X_1 + \frac{\beta_s}{s} - \frac{\mathbb{E}[\beta_s | \mathcal{F}_s^X]}{\sqrt{s(1-s)}} \right) ds = B_t - \int_0^t \left(\frac{\beta_s}{s(1-s)} - \frac{\mathbb{E}[\beta_s | \mathcal{F}_s^X]}{s(1-s)} \right) ds.$$

This expression shows that $\mathbb{E}[|W_t^X|] < \infty$. Moreover, for $0 \leq s < t \leq 1$, $\mathbb{E}[B_t - B_s | \mathcal{F}_s^X] = \mathbb{E}[\mathbb{E}[B_t - B_s | \mathcal{F}_s^\beta \vee \sigma(X_1)] | \mathcal{F}_s^X] = 0$ because β is independent of X_1 and B is a standard $\mathbf{F}^\beta$ -Wiener process in $\mathbb{R}^d$. Therefore,

$$\mathbb{E}[W_t^X - W_s^X | \mathcal{F}_s^X] = \int_s^t \left(\frac{\mathbb{E}[\beta_u | \mathcal{F}_s^X]}{u(1-u)} - \frac{\mathbb{E}[\mathbb{E}[\beta_u | \mathcal{F}_u^X] | \mathcal{F}_s^X]}{u(1-u)} \right) du = 0.$$

Since W^X is continuous and $\mathbf{F}^X$ -adapted by construction, this means that W^X is a continuous $\mathbf{F}^X$ -martingale. Also, $[W^X, W^X]_t = [B, B]_t = tI_d$ for all $t \in [0, 1]$. Hence W^X is a standard $\mathbf{F}^X$ -Wiener process in $\mathbb{R}^d$ by Lévy's characterization. $\square$

A.2 Proof of Proposition 2.3

The proof is an application of Girsanov's theorem. For later use, we formulate it in a general form. Let $w = (w_t)_{t \in [0,1]}$ be the canonical process on $\mathbb{W}^d$. Set $\mathcal{B}_t = \sigma(w_s : s \leq t)$ for $t \in [0, 1]$. For two probability distributions P and Q on $\mathbb{W}^d$ and $t \in [0, 1]$, we write $H_t(P | Q) := H(P|_{\mathcal{B}_t} | Q|_{\mathcal{B}_t})$ for short.

Lemma A.3. *For each $i = 1, 2$, let $b_i : (0, 1) \times \mathbb{W}^d \rightarrow \mathbb{R}^d$ be a progressively measurable functional. Suppose that $X^{(i)} = (X_t^{(i)})_{t \in [0,1]}$ be a weak solution to the following SDE:*

$$dX_t^{(i)} = b_i(t, X^{(i)})dt + dW_t, \quad X_0^{(i)} = 0.$$

Suppose also that for all $t \in [0, 1]$

$$\begin{cases} P^{X^{(1)}} \left(\int_0^t |b_1(s, w)|^2 ds < \infty \right) = P^{X^{(2)}} \left(\int_0^t |b_2(s, w)|^2 ds < \infty \right) = 1, \\ P^W \left(\int_0^t |b_2(s, w)|^2 ds < \infty \right) = 1. \end{cases} \quad (\text{A.3})$$

Then

$$H_t(P^{X^{(1)}} | P^{X^{(2)}}) = \frac{1}{2} \int_0^t \mathbb{E} \left[|b_1(s, X^{(1)}) - b_2(s, X^{(1)})|^2 \right] ds \quad (\text{A.4})$$

for all $t \in [0, 1]$. Moreover, if (A.3) holds with $t = 1$, (A.4) also holds with $t = 1$.

Proof. The second claim can be shown in a similar manner to the first one, so we focus on the proof of the first claim. By Theorem 7.6 in [41], $P^{X^{(1)}}|_{\mathcal{B}_t} \ll P^W|_{\mathcal{B}_t}$ for any $t \in [0, 1)$ and the process $(\frac{dP^{X^{(1)}}|_{\mathcal{B}_t}}{dP^W|_{\mathcal{B}_t}})_{t \in [0, 1)}$ has a continuous modification $(D_t^{(1)})_{t \in [0, 1)}$ with respect to P^W such that

$$D_t^{(1)}(w) = \exp \left(\int_0^t b_1(s, w) dw_s - \frac{1}{2} \int_0^t |b_1(s, w)|^2 ds \right) \quad P^{X^{(1)}}\text{-a.s.}$$

Similarly, by Theorem 7.7 in [41], $P^{X^{(2)}}|_{\mathcal{B}_t} \sim P^W|_{\mathcal{B}_t}$ for any $t \in [0, 1)$ and the process $(\frac{dP^{X^{(2)}}|_{\mathcal{B}_t}}{dP^W|_{\mathcal{B}_t}})_{t \in [0, 1)}$ has a continuous modification $(D_t^{(2)})_{t \in [0, 1)}$ with respect to P^W such that

$$D_t^{(2)}(w) = \exp \left(\int_0^t b_2(s, w) dw_s - \frac{1}{2} \int_0^t |b_2(s, w)|^2 ds \right) \quad P^W\text{-a.s.} \quad (\text{A.5})$$

Since $P^{X^{(1)}}|_{\mathcal{B}_t} \ll P^W|_{\mathcal{B}_t}$, Eq.(A.5) also holds $P^{X^{(1)}}\text{-a.s.}$ Therefore, with $\Lambda_t := D_t^{(1)}/D_t^{(2)}$,

$$\Lambda_t(w) = \exp \left(\int_0^t (b_1(s, w) - b_2(s, w)) dw_s - \frac{1}{2} \int_0^t (|b_1(s, w)|^2 - |b_2(s, w)|^2) ds \right) \quad P^{X^{(1)}}\text{-a.s.}$$

Hence

$$\log \Lambda_t(X^{(1)}) = \frac{1}{2} \int_0^t |b_1(s, X^{(1)}) - b_2(s, X^{(1)})|^2 ds + \int_0^t (b_1(s, X^{(1)}) - b_2(s, X^{(1)})) dW_s \quad \text{a.s.} \quad (\text{A.6})$$

Meanwhile, by the definition of $D^{(1)}$ and $D^{(2)}$,

$$\Lambda_t = \frac{dP^{X^{(1)}}|_{\mathcal{B}_t}}{dP^{X^{(2)}}|_{\mathcal{B}_t}} \quad P^W\text{-a.s. and } P^{X^{(2)}}\text{-a.s.} \quad (\text{A.7})$$

Since $P^{X^{(1)}}|_{\mathcal{B}_t} \ll P^W|_{\mathcal{B}_t}$, this also holds $P^{X^{(1)}}\text{-a.s.}$ Hence

$$H_t(P^{X^{(1)}} | P^{X^{(2)}}) = \mathbb{E}[\log \Lambda_t(X^{(1)})]. \quad (\text{A.8})$$

Now, if the right hand side of (A.4) is finite, then

$$\mathbb{E} \left[\int_0^t (b_1(s, X^{(1)}) - b_2(s, X^{(1)})) dW_s \right] = 0.$$

Hence (A.6) and (A.8) give (A.4). Otherwise, for every $n \in \mathbb{N}$, define a function $\tau_n : \mathbb{W}^d \rightarrow [0, 1]$ as $\tau_n(w) = \inf\{t \in [0, 1] : \int_0^t (|b_1(s, w)|^2 + |b_2(s, w)|^2) ds \geq n\} \wedge 1$ for $w \in \mathbb{W}^d$. We can easily check that τ_n is a $(\mathcal{B}_t)_{t \in [0, 1]}$ -stopping time. Also, from (A.7) we can verify that $(\Lambda_t)_{t \in [0, 1)}$ is a $(\mathcal{B}_t)_{t \in [0, 1)}$ -martingale under $P^{X^{(2)}}$. Hence we have $\Lambda_{t \wedge \tau_n} = \mathbb{E}_{P^{X^{(2)}}}[\Lambda_t | \mathcal{B}_{t \wedge \tau_n}]$ $P^{X^{(2)}}\text{-a.s.}$ for any $t \in [0, 1)$ by the optional sampling theorem. This and (A.7) show that

$$\Lambda_{t \wedge \tau_n} = \frac{dP^{X^{(1)}}|_{\mathcal{B}_{t \wedge \tau_n}}}{dP^{X^{(2)}}|_{\mathcal{B}_{t \wedge \tau_n}}} \quad P^{X^{(2)}}\text{-a.s.} \quad (\text{A.9})$$

Also, since $(0, \infty) \ni x \mapsto x \log x \in \mathbb{R}$ is convex, Jensen's inequality gives

$$\int \Lambda_{t \wedge \tau_n} \log \Lambda_{t \wedge \tau_n} dP^{X^{(2)}} \leq \int \Lambda_t \log \Lambda_t dP^{X^{(2)}}.$$

Combining this with (A.7) and (A.9) gives

$$\mathbb{E}[\log \Lambda_{t \wedge \tau_n}(X^{(1)})] \leq \mathbb{E}[\log \Lambda_t(X^{(1)})]. \quad (\text{A.10})$$

Meanwhile, by (A.6) and the definition of τ_n ,

$$\mathbb{E}[\log \Lambda_{t \wedge \tau_n}(X^{(1)})] = \frac{1}{2} \mathbb{E} \left[\int_0^{t \wedge \tau_n(X^{(1)})} |b_1(s, X^{(1)}) - b_2(s, X^{(1)})|^2 ds \right].$$

Noting $P^{X^{(1)}}|_{\mathcal{B}_t} \ll P^W|_{\mathcal{B}_t}$, we have $\tau_n(X^{(1)}) \uparrow 1$ as $n \rightarrow \infty$ a.s. by (A.3). Hence the monotone convergence theorem gives

$$\lim_{n \rightarrow \infty} \mathbb{E}[\log \Lambda_{t \wedge \tau_n}(X^{(1)})] = \frac{1}{2} \mathbb{E} \left[\int_0^t |b_1(s, X^{(1)}) - b_2(s, X^{(1)})|^2 ds \right] = \infty.$$

Consequently, we have $H_t(P^{X^{(1)}} | P^{X^{(2)}}) = \infty$ by (A.8) and (A.10), which means (A.4). $\square$

Proof of Proposition 2.3. Note that $\mu \ll \gamma_d$ if and only if the law of X is absolutely continuous with respect to P^W . By Theorem 7.5 in [41], this is also equivalent to $\int_0^1 |v_t|^2 dt < \infty$ a.s. Therefore, if $\mu \not\ll \gamma_d$, both sides of (2.1) are infinite and thus (2.1) holds. If $\mu \ll \gamma_d$, Lemma A.3 gives

$$H(P^X | P^W) = \mathbb{E} \left[\frac{1}{2} \int_0^1 |v_s|^2 ds \right].$$

Since $\frac{dP^X}{dP^W}(X) = \frac{d\mu}{d\gamma_d}(X_1)$ by the definition of the Föllmer process, this shows (2.1). $\square$

A.3 Proofs of Proposition 2.8

First, by (2.3), (2.4) and the Markov property of X , we can rewrite v_t as

$$v_t = \frac{\mathbb{E}[X_1 | \mathcal{F}_t^X] - X_t}{1 - t} = \mathbb{E}[X_1 | \mathcal{F}_t^X] - \frac{\mathbb{E}[\beta_t | \mathcal{F}_t^X]}{1 - t}. \quad (\text{A.11})$$

Next, let $B = (B_t)_{t \in [0,1]}$ be the process defined by (A.1). By the proof of Theorem 2.1, B is a standard $\mathbf{F}^\beta$ -Wiener process in $\mathbb{R}^d$. Moreover, solving (A.1) with respect to β , we obtain

$$\beta_t = (1 - t) \int_0^t \frac{1}{1 - s} dB_s$$

for all $t \in [0, 1)$ (cf. [35, Section 5.6.B]). This implies that $(\beta_t/(1 - t))_{t \in [0,1]}$ is an $\mathbf{F}^\beta$ -martingale in $\mathbb{R}^d$. Since β is independent of X_1 and $\mathcal{F}_t^X \subset \mathcal{F}_t^\beta \vee \sigma(X_1)$ for all $t \in [0, 1]$, combining this with the expression (A.11) gives the desired result. $\square$

A.4 Proofs of Proposition 2.9

By Tweedie's formula, we can rewrite v_t as

$$v_t = X_1 + \frac{\beta_t}{t} - \frac{\mathbb{E}[\beta_t | X_t]}{t(1 - t)}.$$

Since $\mathbb{E}[|\beta_t|^r] < \infty$, the asserted claim follows from this expression. $\square$

A.5 Proof of (3.8)

By (A.2), we have

$$\mathbb{E} \left[\nabla \log p_t(X_t) - \frac{-\beta_t}{t(1-t)} \mid X_t \right] = 0.$$

Hence we obtain

$$\mathbb{E} \left[\left(\frac{-\beta_t}{t(1-t)} - \nabla \log p_t(X_t) \right) \cdot (\nabla \log p_t(X_t) - \widehat{\mathbf{s}}(X_t, t)) \right] = 0.$$

Since $(\beta_t/\{t(1-t)\}, X_t)$ has the same law as $(Z/\sqrt{t(1-t)}, \xi_t)$ by Proposition 2.1, we obtain the desired result from the above identity. $\square$

A.6 Proof of Theorem 3.1

The following lemma is a special case of the so-called data processing inequality and well-known in information theory (see e.g. Corollary 2.18 in [50]). We give a direct proof for the sake of readers' convenience.

Lemma A.4. *Let $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{T}, \mathcal{B})$ be two measurable spaces and $T : \mathcal{X} \rightarrow \mathcal{T}$ an $\mathcal{A}/\mathcal{B}$ -measurable function. For any two probability measures P and Q on $(\mathcal{X}, \mathcal{A})$,*

$$H(P^T \mid Q^T) \leq H(P \mid Q),$$

where P^T and Q^T are the image measures of P and Q by T , respectively.

Proof. It suffices to consider the case $H(P \mid Q) < \infty$. In this case, we have $P \ll Q$, so $P^T \ll Q^T$. Moreover,

$$\frac{dP^T}{dQ^T} \circ T = \mathbb{E}_Q \left[\frac{dP}{dQ} \mid T \right] \quad Q\text{-a.s.} \quad (\text{A.12})$$

In fact, the left hand side of (A.12) is evidently $\sigma(T)$ -measurable and for any $B \in \mathcal{B}$

$$\int_{T^{-1}(B)} \left(\frac{dP^T}{dQ^T} \circ T \right) dQ = \int 1_B(y) \frac{dP^T}{dQ^T}(y) Q^T(dy) = P(T^{-1}(B)) = \int_{T^{-1}(B)} \frac{dP}{dQ} dQ.$$

Since $\sigma(T) = \{T^{-1}(B) : B \in \mathcal{B}\}$, this implies (A.12). Hence we obtain

$$\begin{aligned} H(P^T \mid Q^T) &= \int \frac{dP^T}{dQ^T} \log \left(\frac{dP^T}{dQ^T} \right) dQ^T = \int \mathbb{E}_Q \left[\frac{dP}{dQ} \mid T \right] \log \left(\mathbb{E}_Q \left[\frac{dP}{dQ} \mid T \right] \right) dQ \\ &\leq \int \mathbb{E}_Q \left[\frac{dP}{dQ} \log \left(\frac{dP}{dQ} \right) \mid T \right] dQ, \end{aligned}$$

where the last inequality follows by Jensen's inequality and the fact that $x \mapsto x \log x$ is convex. Since the last term is equal to $H(P \mid Q)$, we obtain the desired result. $\square$

Proof of Theorem 3.1. It suffices to consider the case $\mathbb{E}[|X_1|^2] < \infty$. First, note that (2.5) yields $\mathbb{E}[\int_0^t |v_s|^2 ds] < \infty$ for all $t \in [0, 1)$. Next, define a function $b : [0, 1) \times \mathbb{W}^d \rightarrow \mathbb{R}^d$ as

$$b(t, x) = \begin{cases} 0 & \text{if } 0 \leq t < t_0 \text{ or } t_N \leq t < 1, \\ x(t_i)/t_i + \widehat{\mathbf{s}}(x(t_i), t_i) & \text{if } t_i \leq t < t_{i+1} \text{ for some } i = 0, 1, \dots, N-1. \end{cases}$$

Let $\hat{X} = (\hat{X}_t)_{t \in [0,1]}$ be a solution to the following SDE:

$$d\hat{X}_t = b(t, \hat{X})dt + dW_t, \quad \hat{X}_0 = 0.$$

Then $\hat{X}_{t_N}$ has the same law as $X_{t_N}^{\text{EM}}$. Hence, by Lemma A.4

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{EM}}}) \leq H_{t_N}(P^X \mid P^{\hat{X}}).$$

Also, by Lemma A.3,

$$\begin{aligned} H_{t_N}(P^X \mid P^{\hat{X}}) &= \frac{1}{2} \int_0^{t_N} \mathbb{E} [|v_t - b(t, X)|^2] dt \\ &= \frac{1}{2} \int_0^{t_0} \mathbb{E} [|v_t|^2] dt + \frac{1}{2} \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \mathbb{E} \left[\left| v_t - \frac{X_{t_i}}{t_i} - \hat{\mathfrak{s}}(t_i, X_{t_i}) \right|^2 \right] dt. \end{aligned}$$

Therefore, by (3.5),

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{EM}}}) \leq \frac{1}{2} \int_0^{t_0} \mathbb{E} [|v_t|^2] dt + \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \mathbb{E} [|v_t - v_{t_i}|^2] dt + \varepsilon_{\text{score}}^2. \quad (\text{A.13})$$

Since $(v_t)_{t \in (0,1)}$ is an $\mathbf{F}^X$ -martingale in $\mathbb{R}^d$ by Proposition 2.8, we obtain

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{EM}}}) \leq \frac{1}{2} \int_0^{t_0} \mathbb{E} [|v_t|^2] dt + \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} (\mathbb{E} [|v_t|^2] - \mathbb{E} [|v_{t_i}|^2]) dt + \varepsilon_{\text{score}}^2 \quad (\text{A.14})$$

$$\leq \frac{t_0}{2} \mathbb{E} [|v_{t_0}|^2] + \sum_{i=0}^{N-1} h_i (\mathbb{E} [|v_{t_{i+1}}|^2] - \mathbb{E} [|v_{t_i}|^2]) + \varepsilon_{\text{score}}^2. \quad (\text{A.15})$$

Using (3.4), we deduce

$$\begin{aligned} \sum_{i=0}^{N-1} h_i (\mathbb{E} [|v_{t_{i+1}}|^2] - \mathbb{E} [|v_{t_i}|^2]) &\leq \kappa \sum_{i=0}^{N-1} (1 - t_{i+1}) (\mathbb{E} [|v_{t_{i+1}}|^2] - \mathbb{E} [|v_{t_i}|^2]) \\ &= \kappa \sum_{i=1}^N (1 - t_i) \mathbb{E} [|v_{t_i}|^2] - \kappa \sum_{i=0}^{N-1} (1 - t_{i+1}) \mathbb{E} [|v_{t_i}|^2] \\ &= \kappa \sum_{i=0}^{N-1} h_i \mathbb{E} [|v_{t_i}|^2] + \kappa(1 - t_N) \mathbb{E} [|v_{t_N}|^2] - \kappa(1 - t_0) \mathbb{E} [|v_{t_0}|^2] \\ &\leq \kappa \int_{t_0}^{t_N} \mathbb{E} [|v_t|^2] dt + \kappa(1 - t_N) \mathbb{E} [|v_{t_N}|^2]. \end{aligned}$$

Now, by (2.5) and $t_0 \leq 1/2$,

$$\mathbb{E} [|v_{t_0}|^2] \leq d \frac{t_0}{1 - t_0} + \mathbb{E} [|X_1|^2] \leq d + \mathbb{E} [|X_1|^2]$$

and

$$(1 - t_N) \mathbb{E} [|v_{t_N}|^2] \leq d + \delta(\mathbb{E} [|X_1|^2] - d) \leq d + \mathbb{E} [|X_1|^2]$$

and

$$\int_{t_0}^{t_N} \mathbb{E}[|v_t|^2] dt \leq d \int_{t_0}^{t_N} \frac{1}{1-t} dt + \mathbb{E}[|X_1|^2] - d = d \log \left(\frac{1-t_0}{\delta} \right) + \mathbb{E}[|X_1|^2] - d.$$

Consequently,

$$\frac{t_0}{2} \mathbb{E}[|v_{t_0}|^2] + \sum_{i=0}^{N-1} h_i (\mathbb{E}[|v_{t_{i+1}}|^2] - \mathbb{E}[|v_{t_i}|^2]) \leq \frac{t_0}{2} (d + \mathbb{E}[|X_1|^2]) + \kappa d \log \left(\frac{1-t_0}{\delta} \right) + 2\kappa \mathbb{E}[|X_1|^2].$$

Combining this with (A.15) gives the desired result. $\square$

A.7 Proof of Theorem 3.2

The following lemma is equivalent to the exponential decay property of the Fisher information along the Ornstein–Uhlenbeck semigroup (see Eq.(5.7.4) of [2]). We give a direct proof for readers' convenience.

Lemma A.5. *If μ has an almost everywhere differentiable density f with respect to γ_d , then $I(\mathcal{S}_t \mu \mid \gamma_d)/t \leq I(\mu \mid \gamma_d)$ for all $t \in (0, 1)$.*

Proof. We may assume $I(\mu \mid \gamma_d) < \infty$ without loss of generality. Set $g := \sqrt{f}$. Then we have

$$\|g\|_{L^2(\gamma_d)}^2 = \int_{\mathbb{R}^d} f d\gamma_d = 1 < \infty \quad \text{and} \quad \|\nabla g\|_{L^2(\gamma_d)}^2 = \frac{1}{4} \int_{\mathbb{R}^d} \frac{|\nabla f|^2}{f} d\gamma_d = \frac{1}{4} I(\mu \mid \gamma_d) < \infty. \quad (\text{A.16})$$

Therefore, by Proposition 0.2 in the supplement to [23], there exist compactly supported C^∞ functions $g^{(n)} : \mathbb{R}^d \rightarrow \mathbb{R}$ ($n = 1, 2, \dots$) such that $\|g^{(n)} - g\|_{L^2(\gamma_d)} + \|\nabla g^{(n)} - \nabla g\|_{L^2(\gamma_d)} \rightarrow 0$ as $n \rightarrow \infty$.

For each $n \in \mathbb{N}$, define a function $f_t^{(n)} : \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$f_t^{(n)}(y) = \phi_{I_d}(y)^{-1} \int_{\mathbb{R}^d} \phi_{(1-t)I_d}(y - \sqrt{t}x) g^{(n)}(x)^2 \gamma_d(dx) \quad (y \in \mathbb{R}^d).$$

Since $\|g^{(n)} - g\|_{L^2(\gamma_d)} \rightarrow 0$ as $n \rightarrow \infty$, we have $f_t^{(n)}(y) \rightarrow f_t(y)$ and $\nabla f_t^{(n)}(y) \rightarrow \nabla f_t(y)$ as $n \rightarrow \infty$. Hence Fatou's lemma yields

$$I(\mathcal{S}_t \mu \mid \gamma_d) \leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^d} \frac{|\nabla f_t^{(n)}(y)|^2}{f_t^{(n)}(y)} \gamma_d(dy). \quad (\text{A.17})$$

Meanwhile, $f_t^{(n)}(y)$ can be rewritten as

$$f_t^{(n)}(y) = \int_{\mathbb{R}^d} g^{(n)}(x)^2 \phi_{(1-t)I_d}(x - \sqrt{t}y) dx = \int_{\mathbb{R}^d} g^{(n)}(\sqrt{t}y + \sqrt{1-t}z)^2 \gamma_d(dz).$$

Since $g^{(n)}$ is a compactly supported smooth function, we can differentiate under the integral sign on the right hand side and obtain

$$\nabla f_t^{(n)}(y) = \sqrt{t} \int_{\mathbb{R}^d} (2g^{(n)} \nabla g^{(n)})(\sqrt{t}y + \sqrt{1-t}z) \gamma_d(dz).$$

Therefore, by the Schwarz inequality,

$$|\nabla f_t^{(n)}(y)|^2 \leq 4t f_t^{(n)}(y) \int_{\mathbb{R}^d} |\nabla g^{(n)}(\sqrt{t}y + \sqrt{1-t}z)|^2 \gamma_d(dz).$$

Hence

$$\begin{aligned} \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^d} \frac{|\nabla f_t^{(n)}(y)|^2}{f_t^{(n)}(y)} \gamma_d(dy) &\leq 4t \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^d} |\nabla g^{(n)}(z)|^2 \gamma_d(dz) \\ &= 4t \int_{\mathbb{R}^d} |\nabla g(z)|^2 \gamma_d(dz) = tI(\mu \mid \gamma_d), \end{aligned}$$

where the first equality follows by $\|\nabla g^{(n)} - \nabla g\|_{L^2(\gamma_d)} \rightarrow 0$ and the second by (A.16). Combining this with (A.17) gives the desired result. $\square$

The above lemma and de Bruijn's identity (2.2) immediately yield the following corollary:

Corollary A.1 (Gaussian logarithmic Sobolev inequality).

$$H(\mu \mid \gamma_d) \leq \frac{1}{2} I(\mu \mid \gamma_d).$$

Proof of Theorem 3.2. We may assume $I(\mu \mid \gamma_d) < \infty$ without loss of generality. Observe that $\mathbb{E}[\int_0^1 |v_s|^2 ds] < \infty$ by Proposition 2.3 and the Gaussian logarithmic Sobolev inequality. This particularly implies $\mathbb{E}[|X_1|^2] < \infty$ by Proposition 2.9. These facts allow us to obtain

$$H(P^{X_1} \mid P^{X_1^{\text{EM}}}) \leq \frac{1}{2} \int_0^{t_0} \mathbb{E}[|v_t|^2] dt + \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} (\mathbb{E}[|v_t|^2] - \mathbb{E}[|v_{t_i}|^2]) dt + \varepsilon_{\text{score}}^2 \quad (\text{A.18})$$

by a similar argument to the derivation of (A.14). Since $(v_t)_{t \in [0,1]}$ is an $\mathbf{F}^X$ -martingale in $\mathbb{R}^d$ by Proposition 2.8, we obtain

$$\frac{1}{2} \int_0^{t_0} \mathbb{E}[|v_t|^2] dt \leq \frac{t_0}{2(1-t_0)} \int_{t_0}^1 \mathbb{E}[|v_t|^2] dt \leq t_0 H(\mu \mid \gamma_d),$$

where the last inequality follows by Proposition 2.3 and $t_0 \leq 1/2$. Meanwhile, we have $\sup_{t \in [0,1]} \mathbb{E}[|v_t|^2] \leq I(\mu \mid \gamma_d) < \infty$ by Proposition 2.4 and Lemma A.5. Hence $v_1 = \lim_{t \uparrow 1} v_t$ exists a.s. by Doob's convergence theorem. Moreover, $\mathbb{E}[|v_t - v_1|^2] \rightarrow 0$ as $t \uparrow 1$ by [60, Theorem 12.1]. In particular, $(v_t)_{t \in [0,1]}$ is an $\mathbf{F}^X$ -martingale in $\mathbb{R}^d$. Consequently,

$$\begin{aligned} \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} (\mathbb{E}[|v_t|^2] - \mathbb{E}[|v_{t_i}|^2]) dt &\leq \sum_{i=0}^{N-1} h_i (\mathbb{E}[|v_{t_{i+1}}|^2] - \mathbb{E}[|v_{t_i}|^2]) \\ &\leq \bar{h} (\mathbb{E}[|v_1|^2] - \mathbb{E}[|v_0|^2]) \leq \bar{h} \lim_{t \uparrow 1} \mathbb{E}[|v_t|^2] \leq \bar{h} I(\mu \mid \gamma_d), \end{aligned}$$

where the last inequality follows by Lemma A.5. All together, we complete the proof. $\square$

A.8 Proof of Theorem 3.3

It suffices to consider the case $\mathbb{E}[|X_1|^2] < \infty$. Recall that $X_{t_i}^{\text{ada}}$ are obtained via the solution to the SDE (3.12) with $m(t, x)$ replaced by $x/t + (1-t)\widehat{\mathfrak{s}}(x, t)$. Therefore, by an analogous argument to the proof of (A.13) but changing the definition of b to

$$b(t, x) = \begin{cases} -x(t)/(1-t) & \text{if } 0 \leq t < t_0, \\ \frac{x(t_i)/t_i + (1-t_i)\widehat{\mathfrak{s}}(x(t_i), t_i) - x(t)}{1-t} & \text{if } t_i \leq t < t_{i+1} \text{ for some } i = 0, 1, \dots, N-1, \\ 0 & \text{if } t_N \leq t < 1, \end{cases}$$

we obtain

$$\begin{aligned}
H(P^{X_{t_N}} | P^{X_{t_N}^{\text{ada}}}) &\leq \frac{1}{2} \int_0^{t_0} \mathbb{E} \left[\left| v_t + \frac{X_t}{1-t} \right|^2 \right] dt + \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \mathbb{E} \left[\left| v_t - \frac{m(t_i, X_{t_i}) - X_t}{1-t} \right|^2 \right] dt + \varepsilon_{\text{score}}^2 \\
&= \frac{1}{2} \int_0^{t_0} \frac{\mathbb{E}[|m(t, X_t)|^2]}{(1-t)^2} dt + \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \frac{\mathbb{E}[|m(t, X_t) - m(t_i, X_{t_i})|^2]}{(1-t)^2} dt + \varepsilon_{\text{score}}^2 \\
&=: I + II + \varepsilon_{\text{score}}^2,
\end{aligned} \tag{A.19}$$

where the second line follows by (2.3). By Proposition 2.5, Jensen's inequality and $t_0 \leq 1/2$,

$$I = \frac{1}{2} \int_0^{t_0} \frac{\mathbb{E}[\mathbb{E}[X_1 | X_t]^2]}{(1-t)^2} dt \leq \frac{t_0 \mathbb{E}[|X_1|^2]}{2(1-t_0)} \leq t_0 \mathbb{E}[|X_1|^2]. \tag{A.20}$$

Also, by Proposition 2.5 and the Markov property of X ,

$$\begin{aligned}
II &= \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \frac{\mathbb{E}[\mathbb{E}[X_1 | \mathcal{F}_t^X] - \mathbb{E}[X_1 | \mathcal{F}_{t_i}^X]^2]}{(1-t)^2} dt \\
&= \sum_{i=0}^{N-1} \int_{t_i}^{t_{i+1}} \frac{\mathbb{E}[\mathbb{E}[X_1 | \mathcal{F}_t^X]^2] - \mathbb{E}[\mathbb{E}[X_1 | \mathcal{F}_{t_i}^X]^2]}{(1-t)^2} dt \\
&\leq \sum_{i=0}^{N-1} \left(\mathbb{E}[\mathbb{E}[X_1 | \mathcal{F}_{t_{i+1}}^X]^2] - \mathbb{E}[\mathbb{E}[X_1 | \mathcal{F}_{t_i}^X]^2] \right) \frac{h_i}{(1-t_i)(1-t_{i+1})} \\
&= \sum_{i=0}^{N-1} (A_{t_i} - A_{t_{i+1}}) \frac{h_i}{(1-t_i)(1-t_{i+1})},
\end{aligned} \tag{A.21}$$

where $A_t = \mathbb{E}[\text{tr}(\text{Cov}[X_1 | X_t])]$. Hence, by (3.4) and (2.6),

$$\begin{aligned}
II &\leq \kappa \sum_{i=0}^{N-1} (A_{t_i} - A_{t_{i+1}}) \frac{1}{1-t_i} \\
&= \kappa \sum_{i=0}^{N-1} \left(\mathbb{E}[\text{tr}(\Gamma_{t_i})] - \frac{1-t_{i+1}}{1-t_i} \mathbb{E}[\text{tr}(\Gamma_{t_{i+1}})] \right) \\
&= \kappa \sum_{i=0}^{N-1} \frac{h_i}{1-t_i} \mathbb{E}[\text{tr}(\Gamma_{t_{i+1}})] + \kappa \mathbb{E}[\text{tr}(\Gamma_{t_0})] - \kappa \mathbb{E}[\text{tr}(\Gamma_{t_N})] \\
&\leq \kappa \sum_{i=0}^{N-1} \frac{h_i}{1-t_i} \mathbb{E}[\text{tr}(\Gamma_{t_{i+1}})] + \kappa \frac{\mathbb{E}[|X_1|^2]}{1-t_0}.
\end{aligned}$$

Since $t_0 \leq 1/2$, we conclude

$$H(P^{X_{t_N}} | P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + \kappa \sum_{i=0}^{N-1} \frac{h_i}{1-t_i} \mathbb{E}[\text{tr}(\Gamma_{t_{i+1}})] + 2\kappa \mathbb{E}[|X_1|^2] + \varepsilon_{\text{score}}^2.$$

By (2.5), we have for any $t \in (0, 1)$

$$\mathbb{E}[\text{tr}(\Gamma_t)] \leq d + (1-t) (\mathbb{E}[|X_1|^2] - d) = td + (1-t) \mathbb{E}[|X_1|^2].$$

Therefore,

$$\begin{aligned}
& H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{ada}}}) \\
& \leq t_0 \mathbb{E}[|X_1|^2] + \kappa d \sum_{i=0}^{N-1} \frac{h_i t_i}{1-t_i} + \kappa(t_n - t_0) \mathbb{E}[|X_1|^2] + 2\kappa \mathbb{E}[|X_1|^2] + \varepsilon_{\text{score}}^2 \\
& \leq t_0 \mathbb{E}[|X_1|^2] + \kappa d \int_{t_0}^{t_N} \frac{1}{1-t} dt + 3\kappa \mathbb{E}[|X_1|^2] + \varepsilon_{\text{score}}^2 \\
& = t_0 \mathbb{E}[|X_1|^2] + \kappa d \log \left(\frac{1-t_0}{\delta} \right) + 3\kappa \mathbb{E}[|X_1|^2] + \varepsilon_{\text{score}}^2.
\end{aligned}$$

Hence we complete the proof. $\square$

A.9 Proof of Theorem 3.4

The key to the proof is the following fact due to Huang *et al.* [31]:

Lemma A.6 (cf. [31], Lemma 10). *Under the assumptions of Theorem 3.4, there exists a universal constant $C \geq 1$ such that*

$$\mathbb{E}[\text{tr}(\text{Cov}[X_1 \mid X_t])] \leq C \frac{1-t}{t} L$$

for all $t \in (0, 1 - \delta]$.

Since this lemma is stated in a slightly more general form than the original one, we give its proof after showing that Theorem 3.4 follows from this lemma. Here and below, for two real numbers x, y we write $x \lesssim y$ if there exists a universal constant $C > 0$ such that $x \leq Cy$.

Proof of Theorem 3.4. By (A.19) and (A.20), we have

$$H(P^{X_{t_N}} \mid P^{X_{t_N}^{\text{ada}}}) \leq t_0 \mathbb{E}[|X_1|^2] + II + \varepsilon_{\text{score}}^2. \quad (\text{A.22})$$

To bound II , we start from the bound (A.21). Let $i_0 \in \{0, 1, \dots, N-1\}$ be the largest index such that $t_{i_0} \leq 1/2$. By (A.21), we can bound II as

$$\begin{aligned}
II & \leq \sum_{i: 0 \leq i \leq i_0} (A_{t_i} - A_{t_{i+1}}) \frac{h_i}{(1-t_i)(1-t_{i+1})} + \sum_{i: i_0 < i < N} (A_{t_i} - A_{t_{i+1}}) \frac{h_i}{(1-t_i)(1-t_{i+1})} \\
& =: II_1 + II_2.
\end{aligned} \quad (\text{A.23})$$

First we bound II_2 . If $i_0 = N-1$, then $II_2 = 0$. Otherwise, we have $t_{i_0+1} > 1/2$. Hence

$$\begin{aligned}
II_2 & \leq \kappa \sum_{i: i_0 < i < N} (A_{t_i} - A_{t_{i+1}}) \frac{1}{1-t_i} \leq \kappa \left(\frac{A_{t_{i_0+1}}}{1-t_{i_0+1}} + \sum_{i: i_0 < i < N-1} A_{t_{i+1}} \frac{h_i}{(1-t_{i+1})(1-t_i)} \right) \\
& \lesssim \kappa L \left(1 + \sum_{i: i_0 < i < N} \frac{h_i}{1-t_i} \right) \leq \kappa L \left(1 + \log \left(\frac{1-t_0}{\delta} \right) \right),
\end{aligned} \quad (\text{A.24})$$

where the first inequality follows by [A1] and the third by Lemma A.6. Next we bound II_1 . Since $\max_i h_i \leq \kappa$ by [A1] and $A_{t_0} \leq \mathbb{E}[|X_1|^2]$,

$$II_1 \leq 4 \sum_{i:0 \leq i \leq i_0} (A_{t_i} - A_{t_{i+1}}) h_i \leq 4\kappa A_{t_0} \leq 4\kappa \mathbb{E}[|X_1|^2].$$

Combining this with (A.22)–(A.24) gives (3.16). If we further assume (3.17),

$$\begin{aligned} II_1 &\leq 24\kappa \sum_{i:0 \leq i \leq i_0} (A_{t_i} - A_{t_{i+1}}) t_i \leq 24\kappa \left(A_{t_0} t_0 + \sum_{i:0 \leq i < i_0} A_{t_{i+1}} h_i \right) \\ &\lesssim \kappa L \left(1 + \sum_{i:0 \leq i < i_0} \frac{h_i}{t_i} \right) \leq \kappa L (1 + \log(1/t_0)). \end{aligned}$$

Hence we can replace $\kappa \mathbb{E}[|X_1|^2]$ in (3.16) by $C\kappa L \log(1/t_0)$ for some universal constant C . $\square$

Now we turn to the proof of Lemma A.6. The proof strategy is the same as [31, Lemma 10]. We first introduce some notation. We set $m := \mathcal{N}(\mathcal{X}, \varepsilon_0)$. Note that $\log m \leq k \log(1/\varepsilon_0)$ by assumption. Also, by definition we can take an ε_0 -net $\{x_1^*, \dots, x_m^*\}$ for $\mathcal{X}$. Note that $x_1^*, \dots, x_m^*$ are distinct due to the maximality of m . We fix a constant $\rho \in (0, 1/3]$ whose value is specified in the main body of the proof of Lemma A.6. We write $\mathcal{B}_i = \mathcal{B}(x_i^*; \varepsilon_0)$ for $i = 1, \dots, m$ and set

$$\mathcal{I} := \{i \in \{1, \dots, m\} : \mu(\mathcal{B}_i) \geq \rho/m\}.$$

The following three lemmas are counterparts of Lemmas B.3 and B.4 in [31].

Lemma A.7. $P(X_1 \notin \bigcup_{i \in \mathcal{I}} \mathcal{B}_i) \leq \rho$.

Proof. Since $\{\mathcal{B}_1, \dots, \mathcal{B}_m\}$ is a covering of $\mathcal{X}$, $x \in \mathcal{X} \setminus \bigcup_{i \in \mathcal{I}} \mathcal{B}_i$ implies $x \in \bigcup_{i \in \mathcal{I}^c} \mathcal{B}_i$. Hence

$$P\left(X_1 \notin \bigcup_{i \in \mathcal{I}} \mathcal{B}_i\right) \leq P\left(X_1 \in \bigcup_{i \in \mathcal{I}^c} \mathcal{B}_i\right) \leq \sum_{i \in \mathcal{I}^c} P(X_1 \in \mathcal{B}_i) \leq \rho.$$

$\square$

Lemma A.8. Set $Z_t = \beta_t / \sqrt{t(1-t)}$. There exists a universal constant $C_0 \geq 1$ such that

$$|Z_t \cdot (x - y)| \leq C_0 \left(L_\rho \sqrt{\delta} + \sqrt{L_\rho} |x - y| \right) \quad \text{for any } x, y \in \mathcal{X} \quad (\text{A.25})$$

with probability at least $1 - \rho$, where $L_\rho := k \log(1/\varepsilon_0) + \log(1/\rho)$.

Proof. Observe that $Z_t \sim N(0, 1)$. Hence, by the Borell–TIS inequality (see e.g. Proposition A.2.1 in [56]) and (3.14), we have for any $u \geq 0$

$$\begin{aligned} \sup_{x, y \in \mathcal{X}: |x-y| \leq \varepsilon_0} |(x-y) \cdot Z_t| &\leq \mathbb{E} \left[\sup_{x, y \in \mathcal{X}: |x-y| \leq \varepsilon_0} |(x-y) \cdot Z_t| \right] + u \sup_{x, y \in \mathcal{X}: |x-y| \leq \varepsilon_0} \sqrt{\mathbb{E}[|(x-y) \cdot Z_t|^2]} \\ &\leq \sqrt{\delta} k \log(1/\varepsilon_0) + u \varepsilon_0 \end{aligned}$$

with probability at least $1 - 2e^{-u^2/2}$. Taking $u = \sqrt{4\log(1/\rho)}$ gives

$$\sup_{x,y \in \mathcal{X}: |x-y| \leq \varepsilon_0} |(x-y) \cdot Z_t| \leq \sqrt{\delta} k \log(1/\varepsilon_0) + \varepsilon_0 \sqrt{4\log(1/\rho)} \quad (\text{A.26})$$

with probability at least $1 - \rho/2$.

Next, for any $i < j$, we have $\{Z_t \cdot (x_i^* - x_j^*)\}/|x_i^* - x_j^*| \sim N(0, 1)$, so

$$P(|Z_t \cdot (x_i^* - x_j^*)| \geq u|x_i^* - x_j^*|) \leq e^{-u^2/2}$$

for all $u \geq 0$. Thus, setting $u = \sqrt{2\log(m^2/\rho)}$, we obtain by the union bound

$$P(|Z_t \cdot (x_i^* - x_j^*)| \geq u|x_i^* - x_j^*| \text{ for some } 1 \leq i < j \leq m) \leq \binom{m}{2} e^{-u^2/2} \leq \frac{\rho}{2}.$$

Since $u \lesssim \sqrt{L_\rho}$, we have

$$|Z_t \cdot (x_i^* - x_j^*)| \lesssim \sqrt{L_\rho} |x_i^* - x_j^*| \quad \text{for any } 1 \leq i < j \leq m \quad (\text{A.27})$$

with probability at least $1 - \rho/2$.

Finally, for any $x, y \in \mathcal{X}$, there exist $i, j \in \{1, \dots, m\}$ such that $x \in \mathcal{B}_i$ and $y \in \mathcal{B}_j$. Then, on the event that (A.26) and (A.27) jointly holds,

$$\begin{aligned} |Z_t \cdot (x - y)| &\leq |Z_t \cdot (x - x_i^*)| + |Z_t \cdot (x_i^* - x_j^*)| + |Z_t \cdot (x_j^* - y)| \\ &\lesssim \sqrt{\delta} k \log(1/\varepsilon_0) + \varepsilon_0 \sqrt{\log(1/\rho)} + \sqrt{L_\rho} |x_i^* - x_j^*| \\ &\leq \sqrt{\delta} k \log(1/\varepsilon_0) + 3\sqrt{L_\rho} \varepsilon_0 + \sqrt{L_\rho} |x - y| \\ &\leq 4L_\rho \sqrt{\delta} + \sqrt{L_\rho} |x - y|. \end{aligned}$$

This completes the proof. $\square$

Lemma A.9. With C_0 the constant in Lemma A.8, define a random subset $\mathcal{E}_t \subset \mathcal{X}$ as

$$\mathcal{E}_t := \left\{ y \in \mathcal{X} : |y - X_1| \geq 5C_0 \sqrt{L_\rho \frac{1-t}{t}} \right\}.$$

Then, with probability at least $1 - 2\rho$,

$$\mu_{t, X_t}(\mathcal{E}_t) \leq \exp(-3C_0 L_\rho). \quad (\text{A.28})$$

Proof. Let Ω_0 be the event on which $X_1 \in \bigcup_{i \in \mathcal{I}} \mathcal{B}_i$ and (A.25) jointly hold. By Lemmas A.7 and A.8, $P(\Omega_0) \geq 1 - 2\rho$. Hence we complete the proof once we show that (A.28) holds on Ω_0 .

Observe that for every $y \in \mathbb{R}^d$

$$\begin{aligned} \frac{|y|^2}{2} - \frac{|y - X_t|^2}{2(1-t)} &= -\frac{t|y|^2 - 2X_t \cdot y + |X_t|^2}{2(1-t)} = -\frac{t|y - X_1|^2 - 2\beta_t \cdot y - t|X_1|^2 + |X_t|^2}{2(1-t)} \\ &= -\frac{U_t(y) - 2\beta_t \cdot X_1 - t|X_1|^2 + |X_t|^2}{2(1-t)}, \end{aligned}$$

where $U_t(y) := t|y - X_1|^2 - 2\beta_t \cdot (y - X_1)$. Hence we can rewrite $\mu_{t,X_t}(\mathcal{E}_t)$ as

$$\mu_{t,X_t}(\mathcal{E}_t) = \frac{\int_{\mathcal{E}_t} \exp\left(-\frac{U_t(y)}{2(1-t)}\right) \mu(dy)}{\int_{\mathbb{R}^d} \exp\left(-\frac{U_t(y)}{2(1-t)}\right) \mu(dy)}. \quad (\text{A.29})$$

Now, if $y \in \mathcal{E}_t$, we have on Ω_0

$$\begin{aligned} U_t(y) &\geq t|y - X_1|^2 - 2C_0\sqrt{t(1-t)} \left(L_\rho\sqrt{\delta} + \sqrt{L_\rho}|y - X_1| \right) \quad (\because (\text{A.25})) \\ &= t|y - X_1| \left(|y - X_1| - 2C_0\sqrt{L_\rho \frac{1-t}{t}} \right) - 2C_0\sqrt{t(1-t)}L_\rho\sqrt{\delta} \\ &\geq 3C_0\sqrt{L_\rho t(1-t)}|y - X_1| - 2C_0\sqrt{t(1-t)}L_\rho\sqrt{\delta} \quad (\because y \in \mathcal{E}_t) \\ &\geq C_0\sqrt{L_\rho t(1-t)} \left(3|y - X_1| - 2\sqrt{L_\rho(1-t)} \right) \quad (\because \delta \leq 1-t) \\ &\geq 13C_0L_\rho(1-t) \quad (\because y \in \mathcal{E}_t). \end{aligned}$$

Hence

$$\int_{\mathcal{E}_t} \exp\left(-\frac{U_t(y)}{2(1-t)}\right) \mu(dy) \leq \exp\left(-\frac{13}{2}C_0L_\rho\right). \quad (\text{A.30})$$

Meanwhile, on the event Ω_0 we can take an $i^* \in \mathcal{I}$ so that $X_1 \in \mathcal{B}_{i^*}$. Then, if $y \in \mathcal{B}_{i^*}$,

$$U_t(y) \leq t\varepsilon_0^2 + 2\sqrt{t(1-t)} \cdot C_0 \left(L_\rho\sqrt{\delta} + \sqrt{L_\rho}\varepsilon_0 \right) \leq 5C_0L_\rho(1-t),$$

where the first inequality follows by (A.25) and the second by $\varepsilon_0 \leq \sqrt{L_\rho\delta}$ and $\delta \leq 1-t$. Further, since $i^* \in \mathcal{I}$,

$$\mu(\mathcal{B}_{i^*}) \geq \exp(-\log(m/\rho)) \geq \exp(-L_\rho) \geq \exp(-C_0L_\rho).$$

Consequently,

$$\begin{aligned} \int_{\mathbb{R}^d} \exp\left(-\frac{U_t(y)}{2(1-t)}\right) \mu(dy) &\geq \int_{\mathcal{B}_{i^*}} \exp\left(-\frac{U_t(y)}{2(1-t)}\right) \mu(dy) \\ &\geq \mu(\mathcal{B}_{i^*}) \exp\left(-\frac{5C_0L_\rho}{2}\right) \geq \exp\left(-\frac{7}{2}C_0L_\rho\right). \end{aligned} \quad (\text{A.31})$$

Combining (A.29)–(A.31) gives (A.28) on Ω_0 . □

Proof of Lemma A.6. Define the random set $\mathcal{E}_t$ in Lemma A.9 with $\rho = \delta^2/R^4$. Note that $L_\rho \lesssim L$ in this case. Also, Lemma A.9 gives

$$\mathbb{E}[\mu_{t,X_t}(\mathcal{E}_t)] \leq \exp(-3C_0L_\rho) + 2\rho \leq 3\rho. \quad (\text{A.32})$$

Now, observe that Proposition 2.5 gives

$$\text{tr}(\text{Cov}[X_1 \mid X_t]) = \int_{\mathbb{R}^d} |y - m(t, X_t)|^2 \mu_{t,X_t}(dy) \leq \int_{\mathbb{R}^d} |y - X_1|^2 \mu_{t,X_t}(dy).$$

Therefore, using the definition of $\mathcal{E}_t$, we deduce

$$\begin{aligned} \text{tr}(\text{Cov}[X_1 \mid X_t]) &\leq \int_{\mathcal{E}_t} |y - X_1|^2 \mu_{t,X_t}(dy) + \int_{\mathcal{E}_t^c} |y - X_1|^2 \mu_{t,X_t}(dy) \\ &\leq 2 \int_{\mathcal{E}_t} |y - \mathbb{E}[X_1]|^2 \mu_{t,X_t}(dy) + 2|X_1 - \mathbb{E}[X_1]|^2 \mu_{t,X_t}(\mathcal{E}_t) + 25C_0^2 \frac{1-t}{t} L_\rho. \end{aligned}$$

By the Schwarz inequality and Proposition 2.5,

$$\begin{aligned} \mathbb{E} \left[\int_{\mathcal{E}_t} |y - \mathbb{E}[X_1]|^2 \mu_{t,X_t}(dy) \right] &\leq \sqrt{\mathbb{E} \left[\int_{\mathbb{R}^d} |y - \mathbb{E}[X_1]|^4 \mu_{t,X_t}(dy) \right] \mathbb{E}[\mu_{t,X_t}(\mathcal{E}_t)]} \\ &= \sqrt{\mathbb{E}[|X_1 - \mathbb{E}[X_1]|^4] \mathbb{E}[\mu_{t,X_t}(\mathcal{E}_t)]} \end{aligned}$$

and

$$\mathbb{E}[|X_1 - \mathbb{E}[X_1]|^2 \mu_{t,X_t}(\mathcal{E}_t)] \leq \sqrt{\mathbb{E}[|X_1 - \mathbb{E}[X_1]|^4] \mathbb{E}[\mu_{t,X_t}(\mathcal{E}_t)]}.$$

Combining these estimates with (A.32) gives

$$\mathbb{E}[\text{tr}(\text{Cov}[X_1 \mid X_t])] \lesssim R^2 \sqrt{\rho} + \frac{1-t}{t} L = \delta + \frac{1-t}{t} L \leq \frac{1-t}{t} L.$$

Hence we obtain the desired result. $\square$

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